

STS8 Logistics: A Data Governance Framework

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STS8 Logistics (Strategic Transport Solutions – Class 8) is a third-party logistics (3PL) provider that specializes in Class 8 freight transportation services throughout the United States. The company supports customers across several industries (retail, automotive, food and beverage, electronics, and consumer goods). STS8 manages a large operational network consisting of drivers, trucks, trailers, facilities, and delivery routes to coordinate shipments and maintain transportation efficiency. Because the company operates across multiple regions, STS8 generates large volumes of data that must be managed effectively to ensure accurate reporting and informed decision-making. The data environment for STS8 Logistics is modeled using the publicly available *Logistics Operations Database* (2025).

Objective

The purpose of this project is to develop a data governance strategy for STS8 Logistics that improves operational visibility and supports better organizational decision-making. As a national 3PL provider, STS8 relies on accurate information to manage freight operations, monitor transportation activity, and evaluate operational performance across its network. Without a centralized governance structure, the reporting process can become inconsistent and make it more difficult for leadership to identify operational trends or respond to performance issues efficiently. This project will examine how STS8 can improve the management of transportation, maintenance, safety, and customer information through a centralized database and business intelligence (BI) tools. It will evaluate how stronger governance can enhance data accuracy and operational efficiency to support future operational growth. Additionally, the plan will explore how standardized practices and clearer data management responsibilities can improve accountability across departments and create a more reliable reporting environment for decision-making. The project will also establish measurable governance and operational performance indicators that provide leadership with greater visibility into data quality, reporting effectiveness, and continuous improvement efforts.

Background

Because STS8 operates across multiple regions and business functions, the organization relies heavily on operational data for its daily operations. As the company continues to generate larger amounts of information, governance will become an important part of maintaining effective information systems and supporting organizational decision-making. Data governance refers to the overall management of organizational data through policies and standards that help ensure information is accurate, reliable, secure, and accessible (Colorado State University Global [CSU], 2026).

Data governance supports reporting and implementation efforts by helping STS8 maintain greater consistency and interpretability across systems and business units. As the organization expands its use of BI tools, strong governance practices will provide a foundation for improving reliable reporting and visibility. Some of the reports that STS8 uses relate to monitoring transportation activities, shipment status information, costs, and fleet management. Data management helps ensure that this data can be organized and interpreted in a consistent manner (Gils, 2020). This allows leadership to better evaluate performance and make better judgments. Governance also plays a role in supporting security across the company. STS8 manages large amounts of sensitive information throughout its environment, which makes protecting it a pivotal part of daily operations. Without proper standards, the information may become vulnerable to unauthorized access and inconsistent handling. Effective governance helps

organizations maintain greater control over how information is accessed, managed, and shared across operational systems.

Problem Statement

STS8 Logistics has continued to expand its operations, and the company has faced growing challenges in managing their records. Operational data is generated across multiple systems and departments, making it more difficult to maintain consistency throughout the organization. This has created delays when reviewing performance, monitoring costs, and identifying trends. Differences in how data is entered, managed, and interpreted also reduce confidence in the accuracy of their data. As STS8 increases its use of centralized technology and BI tools, they will be required to design a more structured approach to managing data across the business. However, as their data environment develops, STS8 must ensure that sensitive data remains properly protected across its systems.

Literature Review

Data Governance Framework

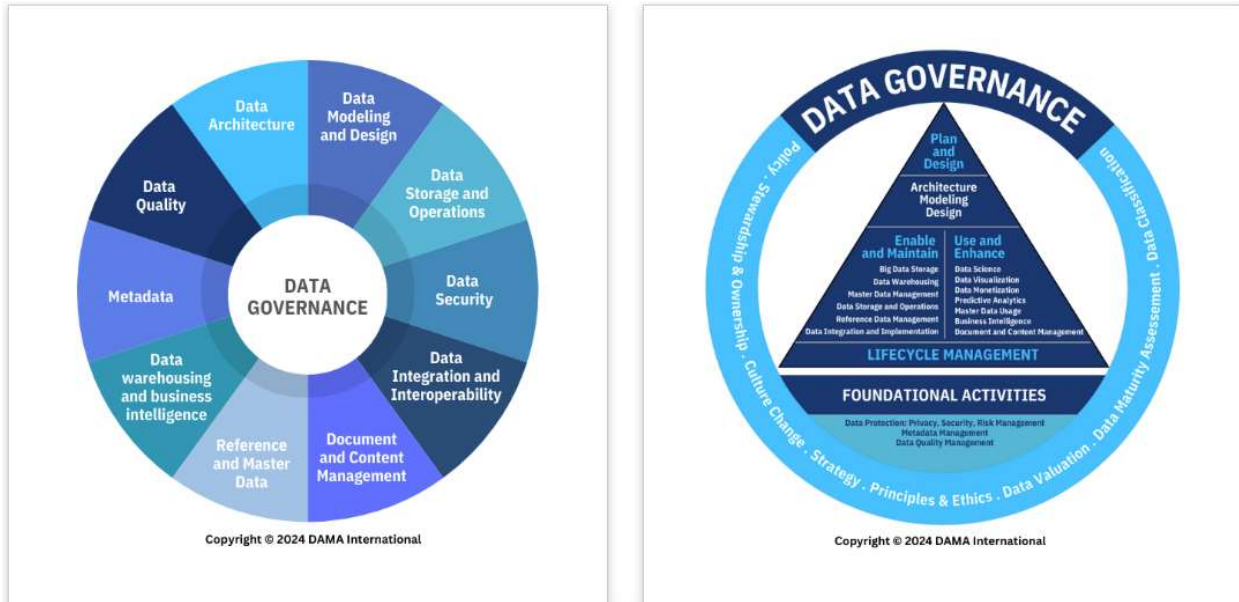
Data governance frameworks provide organizations with structured approaches for managing organizational data through policies, standards, and accountability measures (Gils, 2020). They help establish consistency and reduce errors in data management practices when larger amounts of information are being generated across several departments and systems. Effective governance supports better decision making by improving the quality, reliability, and usability of their data. Frameworks also define responsibilities and establishes oversight that help maintain accountability throughout the data lifecycle.

One of the most widely recognized governance models is the Data Management Body of Knowledge (DAMA-DMBOK). The framework identifies multiple areas of data management, including data governance, data quality, metadata management, security management, master data management, and data architecture (DAMA International, 2024). Unlike governance approaches that focus primarily on technical controls or compliance requirements, DAMA emphasizes the interaction between people, processes, and technology to create an integrated data management environment. As shown in Figure 1, the DAMA-DMBOK framework positions data governance as a central function that supports multiple interconnected domains across an organization. Rather than treating governance as a separate activity, the framework recognizes that data quality, security, architecture, and business intelligence activities influence one another and collectively support organizational objectives.

In addition to defining management domains, DAMA supports organizational policies that establish standards for data creation, storage, access, and maintenance. These policies help organizations maintain data accuracy, completeness, and consistency by creating standardized processes across departments (DAMA International, 2024). The framework also emphasizes clearly defined stewardship responsibilities so that individuals and departments remain accountable for maintaining data quality. These stewardship responsibilities create formal ownership structures that improve accountability and provide oversight for governance activities across organizational systems.

Figure 1

DAMA-DMBOK2 Revised Data Governance Framework



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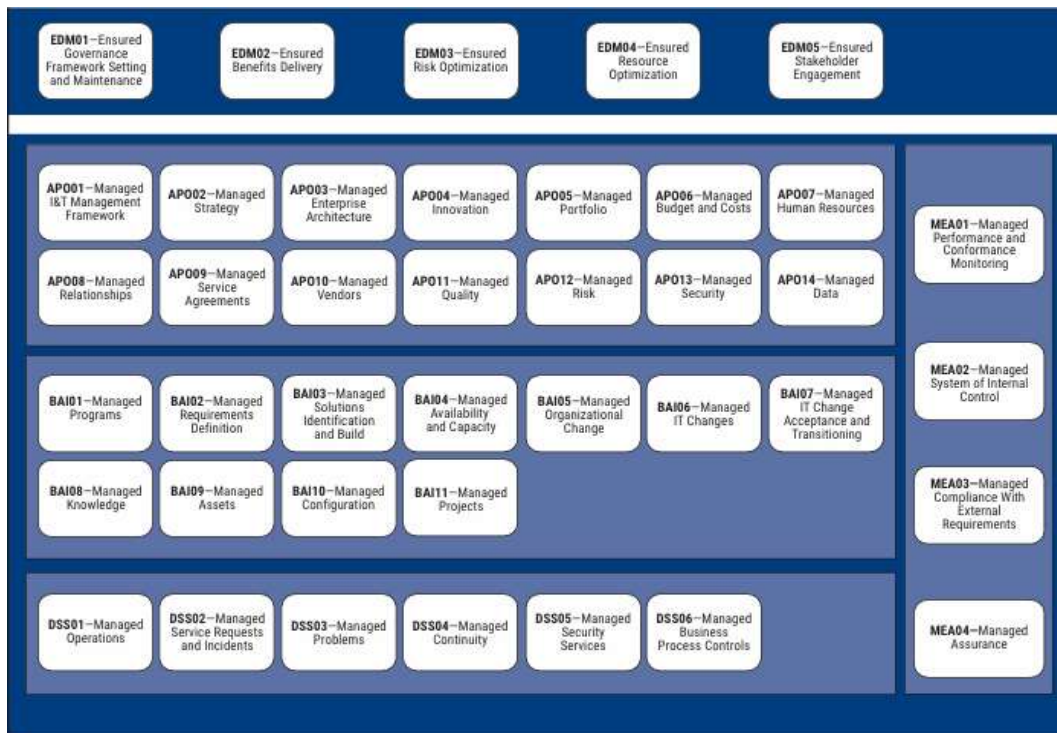
One of the primary strengths of the DAMA framework is its broad and adaptable structure. Organizations operating across multiple departments frequently benefit from the framework because it provides guidance for integrating governance practices across both operational and analytical systems. The framework can also be customized according to organizational needs, allowing companies to focus on specific areas of data management. For organizations experiencing rapid growth or increasing data complexity, this flexibility can support long-term scalability.

However, the same characteristics that make DAMA comprehensive can also create implementation challenges. Because the framework spans numerous management areas, organizations may experience difficulty determining where governance efforts should begin. Smaller organizations or organizations with limited governance maturity may find the framework complex during early implementation stages because it requires coordination across multiple departments and stakeholders. While DAMA provides substantial guidance regarding governance concepts and management domains, it does not prescribe a detailed implementation sequence. This can create uncertainty for organizations seeking highly structured procedures.

Another commonly used governance model is the Control Objectives for Information and Related Technologies (COBIT) framework. ISACA (2019) described COBIT as a framework designed to align information technology governance with business objectives while improving organizational controls and risk management practices. Unlike DAMA, which focuses broadly on managing organizational data, COBIT places greater emphasis on governance processes, performance measurement, compliance requirements, and risk mitigation. As illustrated in Figure 2, the COBIT model organizes governance activities into structured management domains that support planning, implementation, monitoring, and organizational oversight. The framework establishes defined processes and control objectives that allow organizations to measure performance and monitor governance effectiveness. COBIT establishes governance responsibilities and reporting structures that clearly define accountability among stakeholders and support oversight activities across the organization. This

structure represents one of COBIT's primary strengths because organizations can create measurable standards and identify weaknesses through established controls and performance indicators. COBIT further supports governance through formal control objectives and oversight processes that help organizations monitor compliance with established policies (ISACA, 2019). By implementing measurable controls and performance indicators, organizations can identify inconsistencies or deficiencies within data processes before they affect operational reporting and decision-making.

Figure 2
COBIT 2019 Core Model



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One advantage of COBIT is its stronger emphasis on accountability and measurable controls. Organizations can establish key performance indicators and control objectives that support auditing and regulatory requirements. As a result, organizations operating in highly regulated industries often adopt COBIT because it provides stronger governance oversight. However, because COBIT originated primarily from IT governance practices, it may provide less guidance regarding broader data management areas such as metadata management and master data management.

The Data Governance Institute (DGI) framework takes a different approach by concentrating on organizational ownership and decision-making responsibilities. The framework focuses on defining governance rules, assigning authority, and establishing accountability throughout the organization (Data Governance Institute, n.d.). Organizations beginning governance initiatives may benefit from the DGI approach because it creates a practical foundation for assigning responsibilities and establishing oversight structures before implementing larger governance efforts. Similar to other governance frameworks, DGI supports policies and procedures that improve data quality and consistency throughout the organization. The framework emphasizes assigning ownership responsibilities to ensure that individuals remain accountable for maintaining accurate and reliable information.

Although these frameworks differ in structure and implementation, they each emphasize different aspects of governance and data management. DAMA-DMBOK provides a broad enterprise perspective that integrates multiple management domains, COBIT places greater emphasis on controls and governance oversight, and the DGI framework focuses more heavily on ownership and decision-making responsibilities. Organizations may evaluate frameworks according to operational complexity, reporting requirements, regulatory considerations, and organizational goals when determining which approach best aligns with their needs. Despite these differences, the frameworks share several common principles that support effective governance practices. Each framework establishes policies and standards intended to improve data accuracy, completeness, and consistency while defining roles and responsibilities for managing information assets. Additionally, these frameworks create accountability structures and oversight mechanisms that help organizations monitor governance activities and maintain reliable information throughout the data lifecycle. Collectively, these shared characteristics demonstrate that effective governance extends beyond managing data itself and requires coordinated processes, responsibilities, and controls that support organizational decision-making.

Organizations often evaluate governance frameworks independently; however, they are not mutually exclusive. Many organizations combine elements from multiple frameworks to create a governance program that aligns with their operational needs. For STS8 Logistics, the DAMA-DMBOK framework could serve as the primary governance framework because it provides comprehensive guidance for managing organizational data. Its emphasis on data stewardship and data quality also aligns well with the company's reporting and operational objectives. COBIT could complement this by strengthening governance oversight and establishing measurable controls that help leadership evaluate governance performance. The DGI framework could further support implementation by clearly defining ownership responsibilities and decision-making authority across business units. Together, these frameworks provide complementary guidance that allows STS8 to develop a governance program tailored to its operational environment.

Best Practices

Effective data governance extends beyond implementing formal policies and frameworks and requires governance practices to become integrated throughout the organization. Gils (2020) explained that organizations achieve stronger governance outcomes when governance activities become part of daily operational processes rather than isolated technical initiatives. Best practices provide organizations with structured methods for maintaining data quality, improving reporting reliability, and supporting organizational decision-making. Organizations that successfully implement governance initiatives frequently integrate these practices across multiple business functions to establish consistency and accountability.

One commonly recognized best practice is establishing clearly defined ownership and stewardship responsibilities. Data stewards help maintain the quality, integrity, and usability of organizational information by ensuring that governance policies are consistently followed. Stewardship activities may include maintaining data standards, monitoring quality issues, and supporting compliance with governance requirements (Stair & Reynolds, 2018). Establishing ownership responsibilities also creates accountability structures that clarify who is responsible for maintaining and managing information assets throughout the organization. Organizations frequently establish governance committees and cross-functional teams to ensure that governance responsibilities remain coordinated across departments rather than isolated within information technology functions. Access management creates a challenge here because organizations must balance usability with protection. Employees need

access to information in order to perform their responsibilities, but unrestricted access creates additional exposure and increases security risk. For this reason, many organizations rely on role-based access structures that limit permissions according to operational responsibilities. The National Institute of Standards and Technology (2020) recommended restricting access according to business necessity in order to reduce unnecessary exposure to sensitive information. Ongoing monitoring and periodic access reviews further strengthen accountability because they help organizations identify situations where permissions no longer align with current responsibilities.

Maintaining high-quality information also requires organizations to establish standards that determine how data is entered, stored, and interpreted across systems. As organizations grow, information often moves through multiple applications and departments where the same data may be labeled or categorized differently. Even small differences in naming conventions, classifications, or identifiers can create conflicting outputs and reduce confidence in reporting. Master data and reference data support consistency by maintaining shared definitions and common identifiers throughout organizational systems. Standardized definitions therefore reduce duplicate records and help ensure that similar information retains the same meaning regardless of where it is accessed.

Data quality practices become important for the same reason. Organizations are constantly creating and modifying records, meaning information quality changes over time rather than remaining fixed after implementation. Because of this, many organizations use validation rules and monitoring procedures as part of normal operations instead of waiting for issues to become visible in reporting. Some errors are easy to identify, such as incomplete fields or duplicate records, while others may remain hidden until conflicting information begins appearing across systems. Once employees stop trusting the information available to them, additional time is often spent manually checking results before decisions can be made. Even with strong controls in place, governance efforts lose value if reporting cannot be trusted. Leadership decisions are often based on dashboards, reports, and analytical summaries rather than raw data itself. Reporting systems depend heavily on reliable information sources because weak data quality directly affects the usefulness of analytical outputs (Stair & Reynolds, 2018). This is commonly known as “garbage-in, garbage-out”. Reporting standards and metadata practices help provide additional context by allowing users to understand where information originated and how it was processed before being presented.

These practices support different areas of governance, but they become more effective when implemented together rather than independently. Each best practice contributes to maintaining reliable information. Organizations that consistently apply these practices throughout the organization are generally better positioned to strengthen accountability, improve reporting reliability, and support more informed decision-making.

Case Studies

Organizations frequently implement governance initiatives after fragmented systems, inconsistent reporting, and limited visibility begin affecting operational performance. Reviewing real-world implementations provides insight into how organizations address these challenges and establish stronger governance structures. Successful implementations can also identify practical lessons regarding ownership, standardization, and maintaining reliable information across multiple systems.

One example comes from transportation agencies examined by Green and Lucivero (2018). The study reviewed governance practices among multiple state transportation departments, to understand how these organizations managed information across interconnected systems. Transportation agencies often rely on information from numerous operational environments, making consistency and coordination difficult as systems expand. Similar challenges can emerge in environments where transportation data, operational reporting, and business information originate from multiple sources. The agencies addressed these issues by establishing governance policies, defining ownership responsibilities, and creating formal structures for managing information across systems. These initiatives improved coordination among departments and strengthened confidence in the information used for operational planning and reporting.

A second example involved organizations implementing master data management initiatives to improve information consistency across enterprise systems. Vilminko-Heikkinen & Pekkola (2017) found that organizations commonly experienced reporting and operational challenges because similar information existed across multiple systems with inconsistent definitions and management practices. To improve information quality, organizations implemented governance structures that established stewardship responsibilities, standardized business rules, and strengthened oversight of critical information assets. The implementation reduced inconsistencies and improved confidence in the data being used across the organization. Standardized management processes also reduced manual effort previously required to verify information before reporting and decision-making activities.

Several actionable observations emerge from these examples and directly support environments similar to STS8 Logistics. First, organizations operating across multiple systems benefit from establishing ownership responsibilities so that individuals remain accountable for maintaining information quality. Second, standardized definitions and governance policies reduce inconsistencies when operational data moves between departments and reporting environments. Third, ongoing monitoring and validation processes help organizations identify quality issues before they affect planning and decision-making activities. These findings suggest that governance initiatives become more effective when organizations treat data as a shared organizational asset rather than an isolated technical function.

Methodology

Developing an effective data governance strategy requires more than identifying governance frameworks and best practices. Organizations must also establish a structured approach for gathering information, evaluating evidence, and analyzing findings before implementation decisions can be made. Research design provides the overall framework that guides a project and helps ensure that activities align with the intended objectives. Data collection focuses on identifying and obtaining the information needed to evaluate those objectives. This may include information gathered from existing datasets, academic literature, governance frameworks, and other relevant sources. Finally, data analysis involves examining and interpreting the information collected. It is important to maintain consistency between research design, methodological choices, data collection procedures, and analysis techniques in order to produce credible and useful findings (Saunders et al., 2023).

For STS8 Logistics, establishing a methodology is an important step before implementing governance initiatives because it creates a systematic process for evaluating operational requirements

and reporting needs. A clearly defined methodology also helps ensure that future recommendations are based on evidence gathered through a consistent and structured approach. By identifying how information will be collected and analyzed, STS8 can better evaluate governance opportunities and develop a framework that aligns with organizational objectives and operational requirements.

Research Approach & Design

Research design serves as the foundation for a research project by establishing the overall approach used to investigate a problem and answer research questions. The design influences the types of information that will be collected, the methods used to gather that information, and the techniques used to evaluate findings. Selecting an appropriate research design is important because different approaches produce different types of insights. For example, qualitative methods are often used to explore concepts, experiences, and processes, while quantitative methods focus on identifying patterns and relationships within measurable data. In some situations, combining both approaches can provide a more comprehensive understanding of a problem than relying on a single method alone.

This project will utilize a mixed-methods research design that incorporates both qualitative and quantitative approaches. Mixed-methods research combines multiple forms of data collection and analysis to provide a more comprehensive understanding of a problem than either approach could provide independently (Saunders et al., 2023). This design is appropriate for the STS8 Logistics project because the development of a data governance and reporting strategy requires both the evaluation of existing research and the analysis of organizational data. Qualitative research will support the review of governance frameworks, best practices, and case studies discussed in the literature review, while quantitative analysis will provide insight into the operational data that supports reporting and decision-making activities. Combining these approaches will help ensure that future governance policies, reporting requirements, implementation plans, and evaluation metrics are supported by both established research and organizational evidence.

The qualitative component will focus on evaluating existing data governance frameworks, industry best practices, and case studies of successful governance implementations. Information gathered through the literature review will be used to identify common governance principles related to data quality, security, accountability, stewardship, and reporting. Reviewing multiple frameworks and implementation examples provides insight into how organizations establish governance structures and manage information as a strategic asset. The qualitative analysis will also help identify strengths, limitations, and areas of applicability among various governance approaches. These findings will provide a foundation for developing governance policies, roles and responsibilities, and stewardship practices that align with the operational objectives of STS8 Logistics.

The quantitative component will focus on analyzing operational data contained within the Logistics Operations Database (2025). The dataset includes information related to business operations that support organizational reporting and decision-making. Examining this information will help identify reporting requirements, data quality concerns, and governance needs that may affect the accuracy and reliability of business information. Additionally, quantitative analysis will provide insight into how data moves throughout the organization and how information can be structured to support reporting objectives. These findings will support the development of data governance policies, reporting requirements, dashboard design, and performance metrics.

Neither the literature review nor the operational dataset provides a complete picture on its own. The qualitative analysis explains how organizations approach governance and reporting, while the quantitative analysis provides insight into the environment in which those practices may be applied. Combining both perspectives creates a stronger foundation for the remaining phases of the project.

In addition to utilizing a mixed-methods approach, this project will employ a case study research design. According to Yin (2018), case study research is particularly valuable when examining a contemporary phenomenon within its real-world environment and when the goal is to develop an in-depth understanding of a specific case. This design is appropriate because the project focuses on STS8 Logistics and seeks to address organizational challenges related to data governance and reporting. Rather than examining a broad population of organizations, the project concentrates on understanding the unique operational environment, reporting requirements, and governance needs of STS8 Logistics. The case study design complements the mixed-methods approach by providing a focused context in which both qualitative and quantitative forms of evidence can be evaluated. Therefore, a case study design provides an opportunity to evaluate these factors holistically while developing recommendations that align with the specific objectives of STS8 Logistics.

Data Collection

Data collection is the process of gathering information that will be used to support analysis, evaluate findings, and guide decision-making (Stair & Reynolds, 2019). The effectiveness depends heavily on the quality and relevance of the information collected. Even the most sophisticated analytical techniques provide limited value if they are applied to poor data quality. For this reason, organizations must establish a clear data collection strategy that identifies what information is needed, where it will be obtained, and how it will be used to support project objectives. For STS8 Logistics, a structured data collection approach is important because they seek to evaluate both governance practices and operational reporting needs. Collecting information from multiple sources provides a more complete understanding of the organization's environment as well as reduces the risk of making recommendations based on a limited perspective. Establishing a data collection strategy helps ensure that information gathered throughout the research process remains relevant, reliable, and aligned with the overall objectives of the proposed data governance and reporting initiative.

Information will be collected from a combination of secondary and organizational data sources. Secondary sources will include academic literature, governance frameworks, industry publications, and case studies. These sources provide insight into established governance practices, common implementation challenges, and approaches that have been successfully applied in other organizations. Organizational data will primarily consist of operational information contained within the Logistics Operations Database (2025). Examining both research-based and organizational sources helps ensure that recommendations are grounded in established governance principles and remain relevant to the operational needs of STS8 Logistics.

Qualitative data collection methods can provide insight into both governance challenges and broader business concerns that may not be visible through operational data alone. Surveys conducted through Google Forms can be used to gather feedback from employees regarding data quality, reporting effectiveness, process efficiency, and overall confidence in organizational information. Participants will be selected from departments that regularly rely on organizational data. Selecting participants from multiple functional areas provides a broader understanding of governance practices and reporting needs throughout the organization. Similar surveys may also be distributed to customers to measure customer

satisfaction and overall sentiment toward STS8 Logistics. To improve reliability, survey questions should remain consistent across respondents and participation should remain voluntary to encourage honest and unbiased feedback. In-person or video interviews with internal stakeholders can provide a deeper understanding of operational challenges that may not be captured through transactional data. For example, discussions with operations managers and safety personnel may help identify trends within safety incident reports, while interviews with data owners and stewards can provide insight into reporting needs and operational concerns. Focus groups conducted through periodic workshops can further support collaboration by allowing employees from multiple departments to discuss reporting requirements, data quality concerns, and operational priorities. This setting can be beneficial as it encourages discussion among stakeholders who rely on the same information for different purposes. Information gathered through interviews and focus groups will be reviewed using thematic analysis to identify recurring governance concerns. Coding qualitative responses into common themes allows findings from multiple stakeholders to be compared consistently and integrated with the project's quantitative analysis. These methods provide context that helps explain not only how information is managed within the organization but also how organizational performance is experienced by employees, customers, and other stakeholders.

On the other hand, quantitative data collection efforts will focus on understanding how STS8 currently performs and identifying trends that may influence reporting decisions. Much of this information can be obtained from historical operational data maintained within the organization's ERP and reporting systems. Before analysis occurs, operational records will be reviewed for completeness, consistency, and potential inaccuracies to help ensure that reporting and governance decisions are based on reliable information. This review should also identify missing values, duplicate records, inconsistent field definitions, and potential outliers that could affect the accuracy of reporting or analytical results. Missing or incomplete records should be evaluated and addressed before analysis to minimize their impact on reporting accuracy. Duplicate records and inconsistent data definitions should be resolved through standardized data cleansing procedures so that information is interpreted consistently throughout the organization. Performing these data preparation activities before analysis improves the reliability of analytical findings and supports the governance objective of maintaining trustworthy organizational data.

Rather than providing a single snapshot of performance, historical records allow managers to evaluate changes over time and determine whether operational improvements are producing the intended results. For example, delivery performance, safety outcomes, and customer activity can be tracked across multiple reporting periods to identify recurring patterns that may warrant additional investigation. However, only utilizing internal datasets does not always provide enough context for decision-making. An organization may observe improving performance and still lag behind industry standards, or it may identify declining performance without understanding the magnitude of the issue. For this reason, STS8 must supplement internal reporting with industry benchmarks obtained through transportation associations, government transportation agencies, and market research publications. Comparing organizational performance against external benchmarks provides additional perspective and helps leadership determine whether observed results reflect industry trends or issues specific to the organization. Where possible, findings should be validated across multiple sources to strengthen confidence in the conclusions drawn from the analysis. Together, these quantitative sources provide objective evidence that can support governance decisions and long-term performance measurement.

Data Analysis

Data analysis is the process of examining information to identify meaningful insights that support decision-making. While data collection focuses on gathering relevant information, data analysis transforms that information into knowledge that can be used to address organizational challenges and improve performance (Stair & Reynolds, 2018). For STS8, it will be used to evaluate governance requirements, reporting needs, security, and operational performance in order to support the development of future governance and reporting initiatives.

A significant portion of the analysis will focus on understanding operational performance and identifying trends that may influence reporting decisions. Historical data obtained from organizational systems can be examined using descriptive statistics to summarize key performance measures and identify areas that may require additional attention. Measures (averages, frequencies, percentages, distributions) can provide insight into operational activity while helping to identify unusual patterns or inconsistencies within the data. Trend analysis will also be used to evaluate changes in performance over time, allowing STS8 to determine whether operational initiatives are producing the intended results. These techniques are appropriate because they simplify large volumes of information into meaningful insights that can support decision-making.

Additional analytical techniques may also be used to further evaluate operational performance and reporting needs. Regression analysis can help identify relationships between operational variables and business outcomes, allowing STS8 to better understand which factors contribute most significantly to organizational performance. For example, regression models could be used to examine how delivery times, facility locations, customer characteristics, or operational activities influence sales performance or other key metrics. Because the organization's data contains both categorical and continuous variables, regression analysis can provide valuable insight into the factors that drive business results. Clustering techniques may also be utilized to identify natural groupings within organizational data. As an unsupervised learning method, clustering can reveal patterns that may not be immediately apparent through traditional reporting. For STS8, clustering could be used to group customers with similar purchasing behaviors, identify facilities with comparable operational characteristics, or categorize transportation routes based on performance metrics. These insights can support more targeted decision-making and help management identify opportunities for process improvement. Statistical analysis will primarily be conducted using SAS due to its ability to efficiently process large datasets and generate descriptive outputs that support business analysis and reporting. All of these analytical techniques mentioned allow STS8 to examine information from multiple perspectives.

Not all organizational information will exist in a structured format. Customer surveys, employee feedback, interview notes, and safety incident narratives may contain valuable insights that are difficult to evaluate through traditional statistical methods alone. For this reason, text mining techniques may be used to analyze qualitative information and identify recurring themes, frequently used terms, and overall sentiment. Basic text analysis and sentiment scoring can be performed using Excel to help identify common concerns, opportunities for improvement, and emerging issues that may warrant additional investigation. Incorporating text mining alongside traditional statistical analysis provides a more comprehensive understanding of organizational performance by combining operational metrics with stakeholder perspectives.

Although statistical analysis and text mining can reveal important trends within organizational data, the findings are only valuable if decision-makers can understand and apply them. Leadership teams rarely work directly with statistical outputs when making operational decisions. Instead, they rely on reports and visualizations that summarize complex information and present it in a format that is

easier to interpret. For STS8, reporting will serve as the bridge between analytical findings and organizational decision-making. Information identified through statistical analysis can be transformed into visual representations that highlight performance trends, operational concerns, and areas requiring management attention. Power BI will be used to support this process by converting analytical results into interactive reporting tools that allow users to explore information and monitor performance over time. Because these reports may contain sensitive information, security considerations must also be incorporated into the reporting environment. Access to dashboards and reports must be governed through role-based permissions to ensure that users only have access to information necessary for their responsibilities. Incorporating security controls into the reporting process helps protect sensitive information while still providing stakeholders with the visibility needed to support decision-making. This approach helps ensure that insights generated through analysis are communicated effectively to the individuals responsible for making decisions.

This methodology provides a structured approach for developing governance recommendations; however, several assumptions and limitations should be acknowledged. The quality of any analysis is ultimately dependent on the quality of the underlying data. Even within a governed environment, operational datasets may contain missing values, duplicate records, inconsistent field definitions, or outdated information that can influence reporting accuracy and analytical results. While the data preparation procedures described earlier are intended to identify and correct many of these issues, some inconsistencies may remain and should be considered when interpreting the findings. For this reason, governance should be viewed as an ongoing process, with periodic data quality assessments conducted to ensure that reporting remains accurate and reliable over time. Operational conditions may also change over time because of seasonal shipping demand, customer growth, regulatory changes, or supply chain disruptions. As a result, governance policies and reporting requirements should be reviewed periodically to ensure they continue to support the organization's operational objectives. Additional limitations exist within the qualitative components of the methodology. Surveys, interviews, and focus groups provide valuable context that cannot always be obtained through transactional data; however, they also rely on participant perspectives and experiences. Individual responses may be influenced by personal bias, incomplete knowledge, or differing interpretations of organizational processes. To reduce these effects, information will be collected from stakeholders across multiple functional areas and validated against operational data whenever possible. Using multiple data sources and analytical methods allows findings to be corroborated from different perspectives, strengthening the overall reliability of the governance recommendations while providing a more comprehensive understanding of STS8's operational environment.

Data Governance Framework

The literature review and methodology established the importance of data governance in supporting organizational reporting, accountability, security, and decision-making. They also identified governance frameworks, best practices, and implementation approaches that organizations use to manage information as a strategic asset. While these concepts provide a foundation for effective governance, STS8 Logistics must ultimately translate them into policies, responsibilities, and stewardship that align with their specific business environment. Governance approaches are not universally applied, as requirements, complexity, needs, and regulatory considerations often influence which practices provide the greatest value (Gils, 2020). As STS8 continues to manage transportation, fleet, customer, and operational data across multiple business functions, governance activities must be designed to support

the organization's operational objectives while maintaining data quality, security, and accountability throughout the information lifecycle.

Data Quality Policy Development

Policy development establishes the standards that guide how information is managed throughout STS8 Logistics. As the organization continues to rely on data to support operational activities and decision-making, clear expectations are needed to ensure information remains reliable and secure. Effective policies also create consistency across departments by defining how data should be handled throughout its lifecycle. The following policies serve as the basis for managing information in a manner that supports operational requirements and long-term governance objectives.

STS8 Logistics will maintain formal data quality standards that govern how organizational information is created, maintained, and used throughout its lifecycle to support reliable reporting and decision-making. Data quality management relies on more than correcting errors after they occur; it requires consistent processes that promote quality throughout the data lifecycle. To promote consistency, STS8 should establish standardized definitions for the information used throughout the organization. Employees in different departments often interact with the same data for different purposes, which can lead to variations in how information is recorded or interpreted. Those differences can create reporting discrepancies and reduce confidence in organizational data. Standards and frameworks play an important role in managing data quality because they provide a structured approach for ensuring information remains suitable for its intended use (Miller et al., 2025). Standardization should be supported through validation controls that identify potential issues before information becomes part of operational processes or reporting activities. Validation requirements may include mandatory fields, format restrictions, and automated verification checks that prevent incomplete or invalid records from being entered into organizational systems. Implementing these controls at the point of entry helps reduce the introduction of errors and improves the overall reliability of organizational data. Rather than relying solely on corrective actions after issues are discovered, validation controls provide a proactive approach to maintaining data quality throughout the information lifecycle.

The policy will also establish standards for managing master data across the organization. Information related to customers, drivers, vehicles, and facilities is frequently used by multiple departments and often serves as the foundation for reporting and operational activities. When these records are maintained separately or updated inconsistently, conflicting information can emerge across systems. Master data provides common identifiers for important business entities, helping organizations maintain consistency across business processes (Gils, 2020). By establishing a centralized approach to managing master data, STS8 can reduce duplication and improve the reliability of information used throughout the organization. This is only half the equation. While master data helps maintain consistency for key business entities, STS8 must also ensure that information is categorized and interpreted consistently throughout the organization. Reference data provides the common values and definitions that support this objective by establishing standardized classifications used across business systems (Gils, 2020). Maintaining these standards can improve communication between departments and reduce discrepancies in operational and management reporting.

STS8 Logistics will conduct ongoing data quality reviews to ensure organizational information remains accurate, complete, and consistent throughout its lifecycle. Miller et al. (2025) emphasizes the need for ongoing evaluation and improvement activities as part of effective data quality management. Periodic reviews of critical datasets can help STS8 identify issues that may affect reporting reliability. These reviews should evaluate whether organizational standards are being followed, verify that required information is being captured consistently, and identify trends that may indicate broader process issues. When issues are identified, data owners and stewards should investigate the underlying cause and coordinate corrective actions to prevent similar issues from recurring. The results of these reviews should also be used to refine existing standards and validation controls, allowing the organization to continuously improve the quality of information used to support operational and strategic decision-making.

Data Security Policy Development

In addition to maintaining data quality, STS8 must establish policies that protect organizational information from unauthorized access and misuse. Data governance extends beyond the management of information quality and includes safeguarding data assets throughout their lifecycle. Access to organizational information will be granted according to business responsibilities and limited to the information necessary for employees to perform their assigned duties. Role-based access controls support this objective by aligning permissions with job requirements while reducing unnecessary exposure to sensitive information. This approach promotes accountability and helps ensure that access decisions remain consistent with governance objectives (DAMA International, 2024). Access permissions should not remain static after they are initially assigned. Employee transfers, role scope changes, and workforce turnover can create situations where users retain access that is no longer necessary for their position. Periodic access reviews provide an opportunity to verify that permissions remain aligned with current responsibilities and identify situations where access should be modified or removed (National Institute of Standards and Technology [NIST], 2020). Regular reviews strengthen oversight and help ensure that organizational controls continue to reflect legitimate business needs.

STS8 will protect organizational information during storage, transmission, and access by implementing authentication, encryption, and other approved security controls. Authentication controls help verify that users are who they claim to be before access is granted to organizational systems. Multi-factor authentication strengthens this process by requiring additional verification beyond a username and password. Sensitive information should also be protected through encryption to reduce the likelihood that data can be viewed or altered if systems or communications are compromised. These controls are widely recognized as effective measures for reducing cybersecurity risk and protecting organizational information assets (NIST, 2020).

Similar to data quality policies, security controls will be monitored continuously. Ongoing monitoring helps organizations identify potential vulnerabilities, verify that security controls are functioning as intended, and detect unauthorized activity before it results in significant operational or organizational impacts. Regular reviews of system access, authentication activity, and security events provide valuable insight into the effectiveness of existing controls and highlight areas that may require

additional attention. Continuous monitoring is an important component of cybersecurity risk management because it supports timely detection and response to emerging threats. Despite preventative controls and ongoing monitoring efforts, security incidents may still occur and should be addressed through established response procedures. A formal incident response process helps ensure that potential threats are identified, investigated, and resolved in a timely manner. NIST (2020) emphasizes the importance of incident response planning as part of a broader cybersecurity strategy because timely detection and response can reduce operational disruption and limit the impact of security events. Documenting incidents and corrective actions also provides information that can be used to strengthen existing controls and improve future security practices.

Data Compliance Policy Development

Data compliance policies help ensure that organizational information is managed according to established requirements and that governance controls are operating as intended. While data quality policies focus on reliability and security policies focus on protection, compliance policies provide a mechanism for verifying that those standards are followed consistently throughout the organization. Effective compliance practices require documented procedures, ongoing oversight, and the ability to demonstrate that governance requirements are being met (Smallwood, 2014). Compliance activities also help identify situations where existing policies may require revision due to changing business needs or regulatory expectations.

To support these objectives, STS8 Logistics will maintain formal record retention and documentation requirements for operational, financial, customer, and governance records. Operational records, maintenance histories, driver information, customer transactions, and financial data often support reporting activities and may be subject to organizational or regulatory requirements. Maintaining consistent retention practices helps ensure that information remains available when needed while reducing the risk associated with incomplete or improperly managed records (Smallwood, 2014). Documentation should also be maintained for governance decisions, policy changes, and corrective actions to provide transparency and support future reviews. STS8 will conduct periodic compliance reviews to verify adherence to governance policies and identify opportunities for continuous improvement. These reviews provide an opportunity to verify that established policies are being followed and that controls continue to support organizational objectives. Areas such as data retention practices, access management procedures, and security controls should be assessed to identify potential gaps or opportunities for improvement. The results of these reviews can be used to strengthen governance practices and support continuous improvement efforts throughout the organization. Auditability is an important component of governance because organizations must be able to verify and demonstrate compliance with established requirements (Smallwood, 2014). Employees should understand the policies that govern those activities. Providing guidance and ongoing communication regarding compliance requirements can help promote consistent policy adherence across departments. Establishing clear expectations also strengthens accountability by ensuring that employees understand their responsibilities for maintaining compliance with organizational standards.

Roles & Responsibilities

Clearly defined roles and responsibilities are necessary to ensure that governance policies are applied consistently throughout STS8 Logistics. Assigning responsibility for governance activities helps ensure that data quality, security, and compliance requirements remain aligned with organizational objectives and support operational decision-making (Stair & Reynolds, 2018). Although governance requires clear accountability, effective data management is rarely the responsibility of a single individual or department. Data frequently moves across multiple business functions, creating situations where responsibilities overlap and require coordination among operational, technical, and leadership stakeholders. Governance activities often involve multiple participants with shared responsibilities for managing and maintaining organizational information (Gils, 2020). Establishing clear ownership while recognizing these interdependencies helps ensure governance activities remain coordinated throughout the organization. Table 1 provides an overview of the primary governance roles within STS8 Logistics, their responsibilities, decision-making authority, and key governance activities. The narrative that follows provides additional detail regarding how each role contributes to the organization's overall governance framework.

Table 1
Governance Roles & Responsibilities

Governance Role	Primary Responsibilities	Decision Authority	Key Governance Activities
Executive Leadership	Establish governance strategy, allocate resources, approve organizational policies	Enterprise governance decisions	Approve governance policies, resolve cross-functional issues, monitor governance performance
Data Owners	Oversee business data within their functional areas	Business ownership of organizational data	Approve access requests, define data standards, establish retention requirements
Data Stewards	Maintain data quality and governance standards	Operational stewardship activities	Manage databases, backups, integrations, system availability
Information Technology	Maintain systems and technical infrastructure	Security controls and access enforcement	Manage authentication, monitor security events, conduct access reviews, support incident response
Information Security	Protect organizational information assets	Review and categorize support requests submitted to IT	System issues, training gaps, technical improvement opportunities
Data & Analytics Personnel	Support reporting and analytics environments	Reporting and analytical support	Maintain reporting systems, support data integration, develop dashboards and analytical outputs
All Employees	Follow governance policies during daily operations	Operational data handling	Enter accurate data, report quality issues, comply with governance policies

Note. The responsibilities presented summarize the governance framework for STS8 Logistics.

Executive leadership serves as the ultimate decision-making authority for governance activities within STS8. Leadership is responsible for establishing governance priorities, approving organizational

policies, allocating resources, and resolving issues that affect multiple business functions. These individuals determine the direction of governance initiatives and accept responsibility for governance-related risks. However, executive leadership will often rely on recommendations from technical specialists and business stakeholders when making governance decisions. While leadership provides strategic oversight, technical teams possess the expertise necessary to evaluate specific controls, technologies, and implementation approaches.

Data ownership should be assigned according to business responsibility rather than technical responsibility (Stair & Reynolds, 2018). Individuals who are responsible for the business processes that generate and use information are best positioned to determine how that information should be managed. Within STS8, data ownership may be assigned to leaders responsible for customer, fleet, operational, and financial activities because they possess the greatest understanding of how information supports organizational objectives. On the other hand, data owners are responsible for approving access requirements, establishing retention expectations, and ensuring that governance policies support operational needs. Data stewards may be designated by data owners to assist with governance activities and help coordinate the implementation of organizational standards across business functions.

The information technology department supports governance through several specialized functions. Infrastructure and systems teams are responsible for maintaining the technology platforms that support organizational operations, including system availability, backup processes, and disaster recovery capabilities. Security teams are responsible for implementing and monitoring technical safeguards such as access controls, authentication requirements, security monitoring, and incident response procedures. Data and analytics personnel support governance by maintaining reporting environments, supporting data integration activities, and assisting with the implementation of governance controls that improve data accessibility and reliability. While these groups have distinct responsibilities, they must work collaboratively to support the organization's governance objectives.

Governance ultimately depends on the actions of employees throughout the organization. Policies and technical controls can establish expectations, but their effectiveness is determined by how information is managed during daily operations. Employees interact with organizational data more frequently than any other group and often represent the first opportunity to prevent errors, identify concerns, and support established governance requirements. When governance practices become part of routine business activities, the organization is better positioned to maintain reliable information and reduce operational risk. By fostering accountability at every level of the organization, STS8 can promote a culture in which governance is viewed as a shared responsibility rather than an isolated administrative function.

Clearly defined responsibilities establish accountability; however, governance also requires measurable outcomes to determine whether those responsibilities are being fulfilled effectively. Therefore, STS8 will monitor governance performance using indicators that evaluate both data quality and policy compliance. Examples include the number of duplicate customer records identified during quality reviews, unresolved data quality issues, access review completion rates, policy exception counts, and the percentage of critical datasets reviewed according to established governance schedules. Monitoring these measures provides leadership with objective evidence that governance activities are

improving data quality, strengthening accountability, and supporting organizational objectives. The results should also be reviewed periodically to identify trends and guide future improvements to governance policies and stewardship practices.

Data Stewardship

Maintaining high-quality data requires ongoing operational oversight. Data stewardship represents the practical application of governance by focusing on the activities used to manage, maintain, and improve organizational information over time. Data stewardship plays a critical role in supporting governance programs because it is responsible for the day-to-day management of data and helps ensure that organizational standards are consistently applied throughout business operations (Mohd Hanif et al., 2024). For STS8, effective stewardship practices help bridge the gap between governance expectations and operational execution by promoting consistency, improving data quality, and supporting informed decision-making throughout the organization.

Data stewardship begins with ensuring that information is entered and maintained according to established organizational standards. By addressing potential issues at the point of entry, STS8 can reduce the volume of downstream corrections and improve the reliability of information used for operational and strategic decision-making. One of the most effective ways to accomplish this objective is through the use of validation controls that identify potential issues before information becomes part of processes or reporting. Data stewards should work closely with data owners and system administrators to establish validation requirements for critical information such as customer records, fleet assets, shipment information, and financial transactions. These controls may include required fields, format restrictions, standardized naming conventions, and automated verification checks that help prevent incomplete or inaccurate information from entering organizational systems.

Data stewardship should also include ongoing monitoring of data quality. Information that meets organizational standards today may become inaccurate or incomplete over time as processes and systems change. Mohd Hanif et al. (2024) identify data quality monitoring, assessment, and assurance as important components of effective stewardship practices. Stewardship activities should therefore include regular reviews of critical datasets to identify missing values, duplicate records, inconsistencies, and other issues that may affect reporting accuracy. These reviews allow data stewards to identify emerging issues before they affect business operations and provide insight into areas that may require additional attention or process improvements (Miller et al., 2025). When data quality issues are identified, formal correction procedures should be established to ensure that information is updated consistently and accurately. Data cleansing activities should be performed according to established procedures to ensure that corrections are applied consistently throughout the organization. Common stewardship activities may include resolving duplicate records, correcting inaccurate values, updating incomplete information, and removing obsolete data that no longer supports business operations. Before significant changes are made, data stewards should verify the accuracy of replacement information and coordinate with appropriate business stakeholders when necessary. Maintaining documentation of significant corrections can also provide an audit trail that supports future reviews and helps identify recurring data quality issues.

Simply correcting an individual record may resolve an immediate issue, but recurring problems often indicate weaknesses within the underlying process. Data stewards should coordinate with data owners and operational personnel to verify the accuracy of information before corrections are made and document significant changes when appropriate. Validation exceptions should be reviewed periodically to identify recurring issues and determine whether additional controls or process improvements are necessary. This approach helps maintain confidence in organizational information while reducing the likelihood that similar issues will reoccur. Effective stewardship also requires organizations to focus on the root causes of data quality issues rather than addressing only the symptoms. Trends identified through validation reports, quality reviews, and corrective actions can provide valuable insight into weaknesses within existing processes (Miller et al., 2025). For example, recurring errors may indicate that employees require additional training, system controls require modification, or organizational standards are not being applied consistently. By using stewardship activities to identify and address the source of data quality issues, STS8 can continuously improve the quality of information used throughout the organization.

Data Reporting

The previous sections of this project established the importance of data governance in supporting data quality, security, compliance, accountability, and stewardship throughout STS8 Logistics. While these activities help ensure that organizational information remains accurate and reliable, governance alone does not create business value. Information becomes valuable when it can be interpreted and used to support organizational decisions. Data reporting serves as the mechanism that transforms operational data into meaningful information that managers and executives can use to evaluate performance and guide future actions (Camm, 2021). As STS8 continues to expand its transportation network, the volume of information generated across the organization will continue to increase. Shipment activity, fleet operations, maintenance records, customer transactions, and financial information all contribute to STS8's data environment. Without a structured reporting process, much of this information remains isolated within operational systems. Decision-makers may spend significant time gathering data from multiple sources before meaningful analysis can occur. In some cases, important trends may go unnoticed entirely because the information needed to identify them is not readily visible.

A formal reporting environment helps address these challenges by organizing information into a format that supports decision-making. Rather than reviewing individual transactions or operational records, managers can monitor key performance indicators (KPI) that summarize organizational performance. Reporting also provides greater visibility into emerging issues. For example, increasing transportation costs or rising maintenance expenses can be identified more quickly when information is presented through consistent reports and dashboards. Companies that can provide decision-makers with timely, reliable information are often better positioned to respond to changing business conditions and identify opportunities for improvement. According to Weill et al. (2024) for Information Systems Research (2024), organizations operating in the top quartile of "real-time business" maturity achieved 62% higher revenue growth and 97% higher profit margins than organizations in the bottom quartile.

The study identified trusted, accessible data and data-driven decision-making as key characteristics that distinguish these high performers. This reinforces the importance for STS8 to develop reporting environments that transform operational data into actionable insights. An effective reporting framework can improve visibility into transportation operations, support more informed management decisions, and help leadership identify opportunities to improve performance.

Reporting Requirements

Before selecting visualization tools or developing dashboards, STS8 Logistics must first determine what information is needed, who will use that information, and how it will support business goals and objectives. Reporting requirements assists this process by identifying the data, metrics, and analytical outputs necessary to support decision-making. Without clearly defined requirements, reports may overlook critical performance indicators or fail to deliver information in a format that supports decision-making. While this information may originate from the same underlying systems, different stakeholders require different perspectives based on their responsibilities and decision-making authority (Camm, 2021). For example, executive leadership may require summarized performance metrics that support strategic planning, while operational personnel often need more detailed reporting to manage daily activities and address emerging issues. Defining reporting requirements helps ensure that information is presented at the appropriate level of detail and supports the decisions stakeholders are expected to make.

Stakeholder Reporting Needs

Reporting requirements are largely determined by the information needs of the individuals responsible for managing organizational performance and daily operations. Although STS8 Logistics relies on a common data environment, stakeholders use information differently depending on their responsibilities and decision-making authority (Camm, 2021). Consequently, reporting should be designed to provide the appropriate level of detail while ensuring that information remains relevant to the decisions being made. Table 2 summarizes the primary stakeholder groups within STS8, the reporting information they require, and the key performance indicators (KPIs) that support their decision-making responsibilities.

Table 2

Stakeholder Reporting Requirements and Key Performance Indicators

Stakeholder Group	Reporting Focus	Key Performance Indicators (KPI)
Executive Leadership	Strategic Planning & Resource Allocation	Revenue trends, cost per mile, fleet utilization, customer concentration risk
Operations Personnel	Transportation Efficiency & Service Levels	Shipment volume, on-time delivery rates, route performance, driver productivity
Fleet Maintenance Personnel	Preventative Maintenance Planning	Downtime trends, maintenance costs, service frequency, truck utilization

Safety & Compliance Personnel	Risk Reduction & Regulatory Compliance	Safety incidents, violations, claims history
Customer Service Personnel	Customer Support & Service Quality	Shipment status, delivery exceptions, customer activity

Note. The stakeholder groups, reporting priorities, and associated performance measures represent the proposed reporting framework for STS8 Logistics and are intended to illustrate the information needs of each user group.

Executive leadership requires reporting that provides a broad view of organizational performance and supports long-term decision-making. Reporting at this level should focus on summarized metrics that allow leadership to quickly assess overall business health without becoming overwhelmed by operational details. This information helps leadership evaluate whether current operations align with strategic objectives and identify areas requiring additional attention. Because executive decisions are generally focused on long-term planning and organizational performance, reporting is often most valuable when delivered through monthly or quarterly dashboards and summary reports.

Operations require more detailed reporting because they are responsible for overseeing daily transportation activities. Unlike executive leadership, these stakeholders must monitor operational performance in real time and respond to issues that may affect service levels. These reports allow managers to identify operational bottlenecks and implement corrective actions when necessary. Access to timely operational information can help ensure that transportation activities remain efficient and customer commitments are consistently met. As a result, operational reporting should be available daily, with certain performance metrics updated in near real time when possible.

Fleet maintenance personnel and safety professionals rely on reporting that supports risk management and operational continuity. These reports provide insight into the condition and reliability of fleet assets. At the same time, they can help identify trends that may increase operational risk or expose the organization to regulatory concerns. Monitoring these metrics allows stakeholders to proactively address potential issues before they result in costly disruptions or safety-related incidents. Since maintenance and safety concerns can develop quickly, reporting should be updated regularly and reviewed on at least a daily or weekly basis depending on operational requirements.

Lastly, customer service personnel require visibility into shipment activity and customer interactions in order to provide effective support. Reports allow representatives to respond quickly to inquiries and resolve issues before they negatively affect customer relationships. Because customer service personnel often serve as the primary point of contact between STS8 Logistics and its customers, access to accurate and timely information is essential for maintaining service quality and customer activity. These stakeholders often require real-time access to shipment status and delivery information to effectively respond to customer inquiries and service issues.

Although reporting requirements vary among stakeholder groups, each relies on accurate information to support decision-making within their area of responsibility. Aligning reporting requirements with stakeholder needs helps ensure that information is presented at an appropriate level of detail while supporting the broader organizational objectives of STS8 Logistics.

Reporting Goals & Objectives

To ensure that STS8 Logistics' reporting environment aligns with organizational needs and supports strategic decision-making, the company must first establish clear reporting goals and objectives. Establishing these priorities helps ensure that reporting efforts remain focused on providing stakeholders with information that is relevant, actionable, and aligned with overall organizational goals (Valacich & George, 2020). These reporting objectives will focus on improving operational visibility, supporting fleet and transportation performance management, strengthening strategic decision-making, and providing insight into customer service and organizational risk.

Goal 1: Improve Operational Visibility Across the Transportation Network

Effective decision-making depends on having visibility into the activities occurring throughout the transportation network. Reporting should provide stakeholders with timely access to operational information that allows them to evaluate performance, identify emerging issues, and respond appropriately when conditions change.

- **Objective 1.1:** Develop reporting that provides visibility into shipment activity, delivery performance, and route operations across the transportation network.
- **Objective 1.2:** Establish reporting that allows managers to identify operational bottlenecks, monitor service levels, and evaluate transportation efficiency over time.

Goal 2: Improve Fleet Performance and Asset Management

The ability to effectively manage fleet assets directly affects transportation capacity, operating costs, and service reliability. Reporting should provide insight into asset utilization and equipment performance so that resources can be managed more efficiently.

- **Objective 2.1:** Develop reporting that supports the monitoring of fleet utilization, vehicle performance, and equipment availability across the organization.
- **Objective 2.2:** Establish reporting that provides visibility into maintenance activity and equipment downtime to support preventative maintenance planning.

Goal 3: Strengthen Strategic Decision-Making

Leadership requires access to summarized information that provides a clear understanding of organizational operations and supports long-term planning. Reporting should transform operational data into meaningful insights that assist leadership with evaluating organizational outcomes, identifying opportunities for improvement, and allocating resources effectively.

- **Objective 3.1:** Develop executive-level reporting that provides visibility into revenue trends, transportation costs, and overall operational performance.
- **Objective 3.2:** Establish reporting that supports the evaluation of customer activity, growth opportunities, and factors that may affect organizational performance.

Goal 4: Support Customer Service and Risk Management

Maintaining service quality and managing organizational risk require timely access to accurate information. Reporting should help stakeholders identify potential concerns early and provide the information necessary to respond effectively.

- **Objective 4.1:** Develop reporting that improves visibility into shipment status, delivery exceptions, and customer activity to support customer service operations.
- **Objective 4.2:** Establish reporting that provides insight into safety incidents, claims activity, and other factors that may affect operational risk and organizational accountability.

Achieving these reporting objectives requires STS8 to establish clear ownership for each dashboard to ensure reported metrics remain accurate, relevant, and aligned with organizational objectives. Data owners and business stakeholders should be responsible for validating KPI definitions. They should also review reporting accuracy and approve changes to dashboard content as business requirements evolve. The Business Intelligence team should maintain the technical reporting environment and oversee data integrations. They should also monitor dashboard performance to ensure dashboards remain available and continue providing reliable information. Establishing these responsibilities strengthens accountability and helps ensure dashboards continue meeting stakeholder needs over time.

Tools & Technologies

The effectiveness of a reporting environment depends on the tools used to transform that data into actionable insights. While programming languages such as R and Python provide powerful capabilities for statistical analysis, predictive modeling, and data manipulation, they often require specialized technical knowledge to develop and maintain reporting solutions. STS8 Logistics requires a reporting platform that can support a diverse group of stakeholders, ranging from executive leadership to operational personnel, while providing timely access to information that supports decision-making. Business intelligence (BI) platforms are specifically designed to address these needs by integrating data from multiple operational systems and presenting information through interactive dashboards and reports (Edenfield, 2026). These tools allow STS8 to centralize reporting, improve access to information, and reduce the reliance on manual reporting processes. The selected BI solution must support the company's operational and strategic reporting requirements while integrating with the broader business intelligence architecture established for the organization. As a result, reporting solutions were evaluated to determine the platform best suited to support the reporting needs of STS8 Logistics.

Evaluation of Reporting Solutions

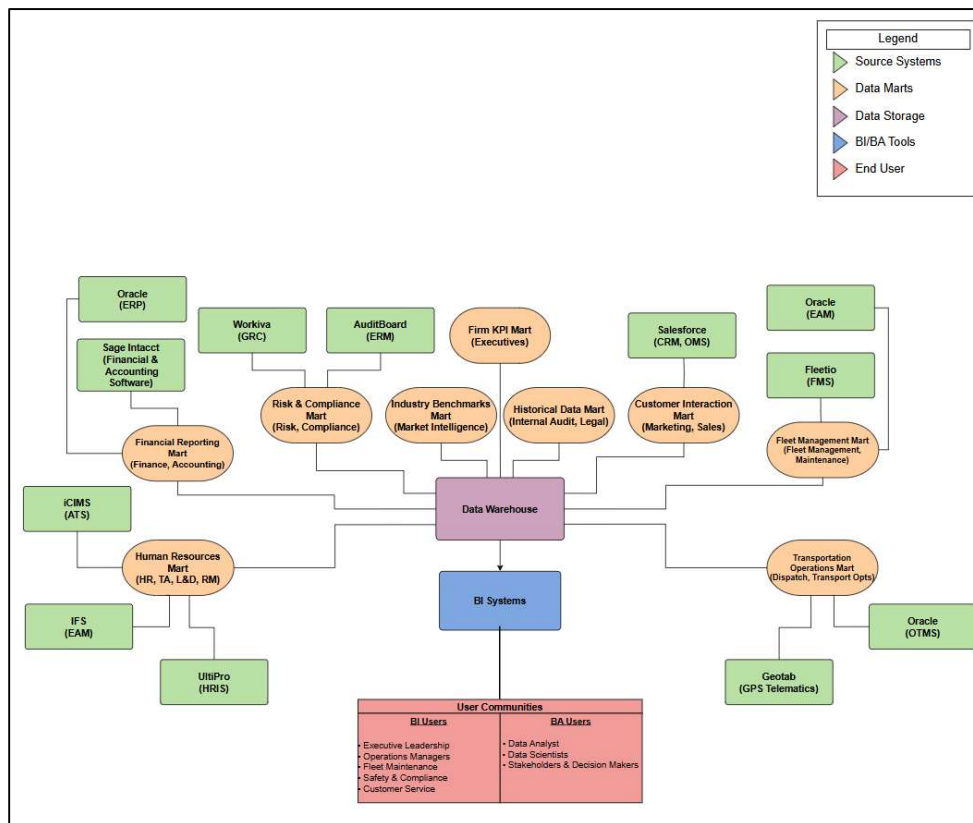
Several business intelligence platforms are capable of supporting the reporting requirements established for STS8 Logistics. Among the most widely adopted solutions are Microsoft Power BI and Tableau. Both platforms provide organizations with the ability to transform operational data into interactive dashboards and visual reports, as well as supports data-driven decision-making across multiple stakeholder groups. However, differences in cost, system integration, scalability, visualization

capabilities, and user experience may influence which platform is best suited for a particular organization.

Figure 3 illustrates the proposed BI architecture for STS8 Logistics. Data originating from operational systems is consolidated within the enterprise data warehouse and organized through subject-specific data marts that support stakeholder reporting requirements. The selected BI platform serves as the interface between the data warehouse and end users, enabling stakeholders to access dashboards, reports, and analytical insights tailored to their responsibilities. Given the central role of the BI platform within the reporting environment, multiple reporting solutions were evaluated to determine the platform best suited to support STS8 Logistics' operational and strategic reporting requirements.

Figure 3

STS8 Logistics System Diagram



Note. Adapted by the author from *Proposal Project: Baker Tilly business intelligence systems architecture* (Schwenk, 2025). Figure was revised to reflect STS8 Logistic's enterprise systems, functional data marts, centralized data storage, reporting tools, and user communities.

Cost and scalability are important considerations when selecting a reporting platform. Power BI is generally recognized as one of the more affordable BI solutions and often provides a lower total cost of ownership than Tableau as organizations expand reporting access across multiple departments (Biswal, 2025). Tableau typically requires a larger licensing investment, particularly when organizations increase the number of users consuming reports and dashboards. However, organizations utilizing Power BI may encounter limitations when sharing reports with users who do not possess the appropriate licensing

requirements (Kfollis et al., 2025). Both platforms are capable of supporting enterprise-scale reporting environments and can effectively manage growing data volumes. For scalability, Tableau utilizes its Hyper data engine to improve performance when working with large datasets (Tableau, n.d.). On the other hand, Power BI leverages Microsoft's cloud infrastructure to support scalability and organizational growth (Biswal, 2025). Consequently, both platforms are capable of supporting STS8 Logistics' future reporting needs, although Power BI may provide a cost advantage as reporting adoption expands.

When deciding on a BI platform, self-service analytics, visualization capabilities, and dashboard functionality are crucial factors to consider. Both Power BI and Tableau allow users to create reports through intuitive interfaces without extensive programming knowledge. Power BI provides a robust collection of visualizations, KPI scorecards, drill-through reports, filters, and interactive dashboards that support both operational and strategic reporting. Tableau is widely recognized as an industry leader in data visualization and storytelling, offering a high degree of customization and flexibility when creating reports and dashboards (Biswal, 2025). Both platforms allow stakeholders to explore information through interactive filtering, drill-down functionality, and dynamic visualizations. However, Tableau is often viewed as the stronger platform for advanced visual analytics, while Power BI provides reporting functionality that satisfies the needs of most business environments.

Data connectivity and system integration are particularly important for STS8 Logistics because reporting requirements span across multiple operational systems. Both platforms support connectivity to databases, spreadsheets, cloud platforms, enterprise applications, and third-party systems. Tableau has historically been recognized for its broad integration capabilities across diverse technology environments, while Power BI offers particularly strong integration within Microsoft's ecosystem (Biswal, 2025). Furthermore, Tableau can leverage the Power BI semantic model, providing organizations with additional flexibility when operating across multiple reporting environments (Buhler, n.d.). However, maintaining multiple BI platforms would likely introduce additional licensing, administration, and training costs. Since STS8 utilizes Oracle technologies throughout several operational systems (ERP, EAM, OTMS), Power BI may provide advantages in integration efficiency, implementation complexity, and long-term connectivity costs.

The last consideration for STS8 when choosing a BI solution is security. The organization manages operational, financial, customer, and compliance-related information that should only be accessible to authorized personnel. Consequently, the selected reporting platform must protect sensitive organizational data while ensuring stakeholders can access the information necessary to perform their responsibilities. Both Power BI and Tableau provide enterprise security capabilities designed to support data governance and organizational accountability. These capabilities include role-based permissions, centralized administration, and governance controls that allow organizations to manage access to reports and dashboards while protecting sensitive information (Biswal, 2025). As reporting adoption expands throughout the organization, these controls become increasingly important for maintaining data integrity and ensuring compliance with organizational policies. Because both platforms provide mature security frameworks capable of supporting enterprise reporting environments, security was not considered a significant differentiator during the evaluation process.

Although both Power BI and Tableau are capable of supporting the reporting requirements established for STS8 Logistics, Power BI was selected as the organization's BI platform. Both solutions provide comparable security controls, self-service analytics capabilities, interactive dashboard functionality, and reporting features that support stakeholder decision-making. Tableau offers stronger data visualization and storytelling capabilities; however, these advantages do not provide sufficient value to offset the higher licensing costs associated with the platform. Because the two solutions are comparable in most functional areas, the decision ultimately centers on cost, scalability, and system integration. Power BI provides a lower total cost of ownership while maintaining the scalability necessary to support future organizational growth. Additionally, STS8's proposed system's environment incorporates several technologies that integrate effectively within Microsoft's ecosystem, making Power BI the more practical solution from both an implementation and long-term maintenance perspective. As a result, Power BI was selected as the reporting platform that best aligns with the operational, technical, and financial requirements of STS8 Logistics.

Selecting an appropriate reporting platform is only one aspect of establishing an effective reporting environment. STS8 must also establish standardized data refresh schedules to ensure dashboards provide stakeholders with timely and reliable information. Operational dashboards that monitor shipment activity, fleet utilization, maintenance performance, and safety incidents should refresh more frequently because they support day-to-day operational decisions. Executive dashboards that summarize organizational performance and long-term trends may require less frequent refresh schedules, such as daily or weekly updates, because they are primarily used for strategic planning. Establishing refresh frequencies according to business requirements helps balance reporting timeliness with system performance while ensuring decision-makers have access to current and consistent information.

Dashboard Design & Reports

The reporting requirements and stakeholder needs identified in previous sections guided the development of a series of dashboards designed to support decision-making throughout STS8 Logistics. Each dashboard was developed to provide stakeholders with information that aligns with their specific responsibilities and reporting requirements. Dashboard design focused on presenting KPIs, trends, and operational metrics through interactive visualizations that allow users to quickly evaluate performance and identify areas requiring attention. To support these objectives, dashboards were developed for customer service personnel, safety and compliance personnel, fleet maintenance personnel, operations, and executive leadership. Each dashboard incorporates visualizations, KPI scorecards, filtering capabilities, and drill-down functionality that enable users to explore information at an appropriate level of detail. The following figures illustrate the proposed reporting environment for STS8 Logistics and demonstrate how Power BI can be used to transform operational data into actionable insights that support both day-to-day management and long-term strategic decision-making.

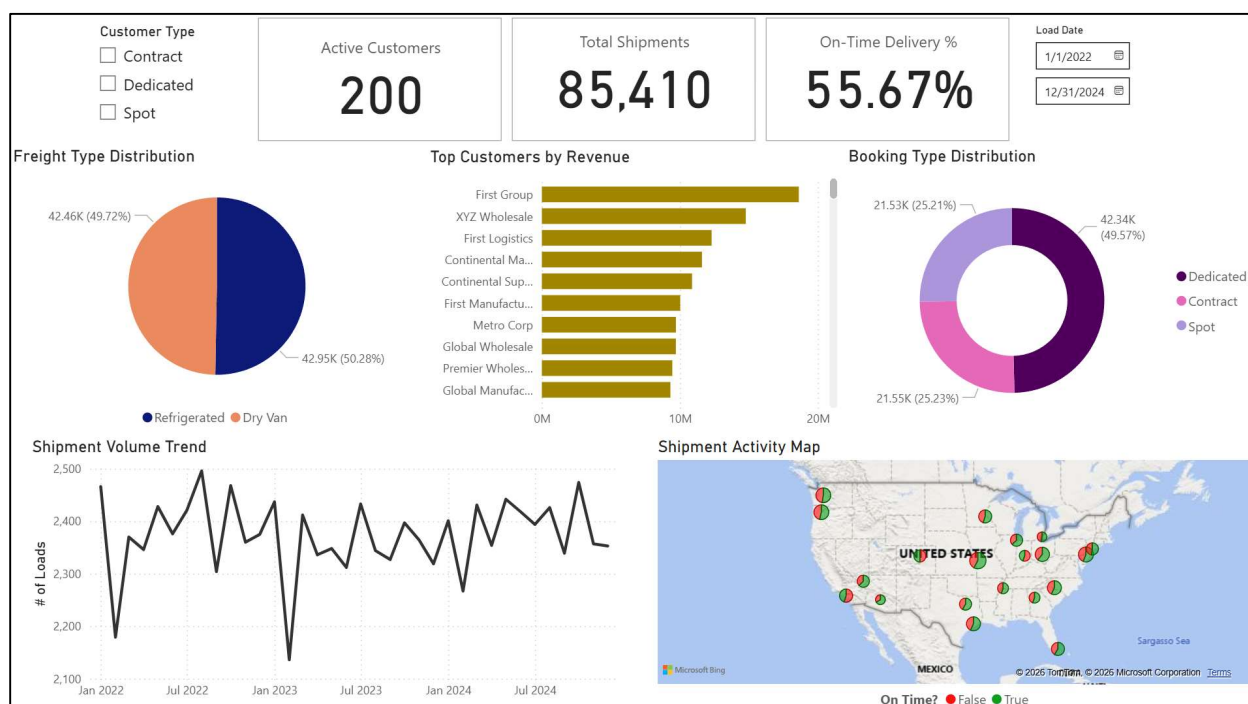
Customer Service Dashboard

The Customer Service Dashboard (Figure 4) was developed to provide customer service personnel with visibility into shipment activity, customer engagement, delivery performance, and freight

movement across the STS8 Logistics network. The dashboard utilizes data from the loads, customers, routes, and delivery_events tables to present operational information related to shipments, customers, load types, revenue generation, and delivery outcomes. Interactive filters allow users to analyze information by customer type and date range, enabling personnel to focus on specific customer segments or reporting periods.

Figure 4

Customer Service Dashboard



Notes. Customer service dashboard displaying shipment activity, customer performance, delivery metrics, and geographic delivery activity for STS8 Logistics.

Customer service personnel can use this dashboard to monitor shipment activity, identify key customers, evaluate delivery performance, and respond to customer inquiries more effectively. The shipment activity map provides visibility into freight movement throughout the transportation network, while the top customers visualization helps personnel understand which customers generate the greatest business value. Delivery performance metrics can be used to identify service issues, investigate delays, and prioritize customer communications. Managers may also use this information to evaluate customer demand patterns and allocate resources to support high-volume customers or geographic markets.

Analysis of the dashboard indicates that STS8 Logistics completed approximately 85,000 shipments during the reporting period while serving 200 active customers. The overall on-time delivery rate was approximately 56%, indicating that nearly half of all shipments experienced some form of delivery delay. Shipment volume remained relatively stable throughout the reporting period, with monthly loads generally ranging between 2,200 and 2,500 shipments. Revenue generation appears concentrated among several key customers, with the largest customer contributing nearly \$20 million in

revenue. Freight activity is distributed across major transportation markets throughout the United States, demonstrating a broad geographic operating footprint. Several patterns are visible within the dashboard. Shipment volumes remained relatively consistent throughout the reporting period with only minor fluctuations, suggesting stable customer demand. Revenue concentration among a relatively small group of customers highlights the importance of maintaining relationships with key accounts while continuing to diversify the customer base. The freight type distribution shows an almost equal split between refrigerated and dry van shipments, indicating balanced demand across both service offerings. The most significant anomaly identified is the relatively low on-time delivery rate compared to industry expectations. This finding may warrant additional investigation into route performance, carrier capacity, scheduling practices, or external factors affecting delivery reliability. Geographic shipment activity appears concentrated within several major transportation markets, providing opportunities to further analyze route profitability and regional service performance. Route-level analysis also revealed significant differences in performance across transportation lanes. Certain routes generated substantially higher revenue than others, while some routes required significantly longer transit times. These findings suggest that STS8 may benefit from further evaluation of route planning, scheduling practices, and resource allocation to improve overall network efficiency.

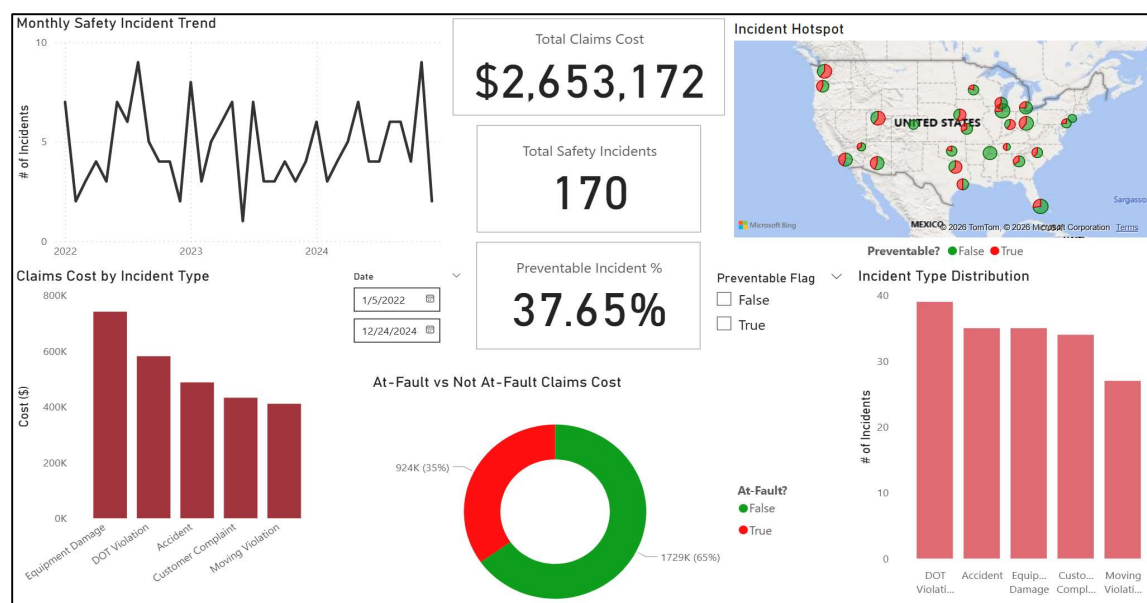
STS8 should prioritize initiatives aimed at improving delivery reliability. Potential actions include route optimization, enhanced dispatch planning, real-time shipment tracking, and performance monitoring of high-delay routes. Improving on-time delivery performance would likely increase customer satisfaction while strengthening competitive positioning within the logistics industry.

Safety & Compliance Dashboard

The Safety & Compliance Dashboard (Figure 5) was developed to provide safety managers, compliance personnel, and operational leadership with visibility into safety incidents, claims activity, regulatory compliance issues, and organizational risk exposure. The dashboard utilizes information from the `safety_incidents` and `claims` datasets to monitor incident frequency, incident types, claims costs, preventable incidents, and geographic incident distribution. Interactive filters allow users to analyze safety performance across different reporting periods and incident classifications, supporting both operational oversight and regulatory compliance efforts.

Figure 5

Safety & Compliance Dashboard



Note. Dashboard developed to support safety and compliance operations through monitoring of safety incidents, claims activity, incident trends, and geographic risk patterns.

Safety and compliance personnel can use this dashboard to monitor incident activity, identify high-risk areas, evaluate claims trends, and prioritize corrective actions. The incident hotspot map provides visibility into locations where incidents occur most frequently, while incident type analysis helps managers identify recurring operational risks. Claims cost metrics can be used to evaluate the financial impact of safety events and support resource allocation decisions related to training, equipment upgrades, and risk mitigation initiatives. Leadership can also use the dashboard to monitor preventable incidents and assess the effectiveness of safety programs over time.

Analysis of the dashboard identified 170 safety incidents during the reporting period, resulting in approximately \$2.65 million in claims costs. Approximately 38% of incidents were classified as preventable, indicating that a substantial portion of safety events may be reduced through process improvements, additional training, or operational controls. Equipment damage incidents generated the highest total claims costs, followed by DOT violations and accidents. The distribution of incidents across multiple categories suggests that safety risks are not concentrated within a single operational area but instead affect several aspects of transportation operations. Geographic analysis further indicates that incidents are distributed throughout the organization's service network rather than being isolated to a specific region. Several patterns emerge from the dashboard analysis. Monthly incident volumes remained relatively consistent throughout the reporting period, generally ranging between two and nine incidents per month, with no clear long-term upward or downward trend. Equipment damage, DOT violations, and accidents represent the most significant contributors to overall claims costs and incident frequency, suggesting these areas should remain priorities for safety improvement efforts. The claims cost analysis also reveals that incident frequency and financial impact do not always align, as certain incident categories generate disproportionately higher costs despite occurring at similar rates. One notable finding is the percentage of preventable incidents, which indicates opportunities to reduce organizational risk through targeted interventions. The geographic distribution of incidents may warrant

additional route-level or facility-level analysis to determine whether localized operational conditions contribute to recurring safety events.

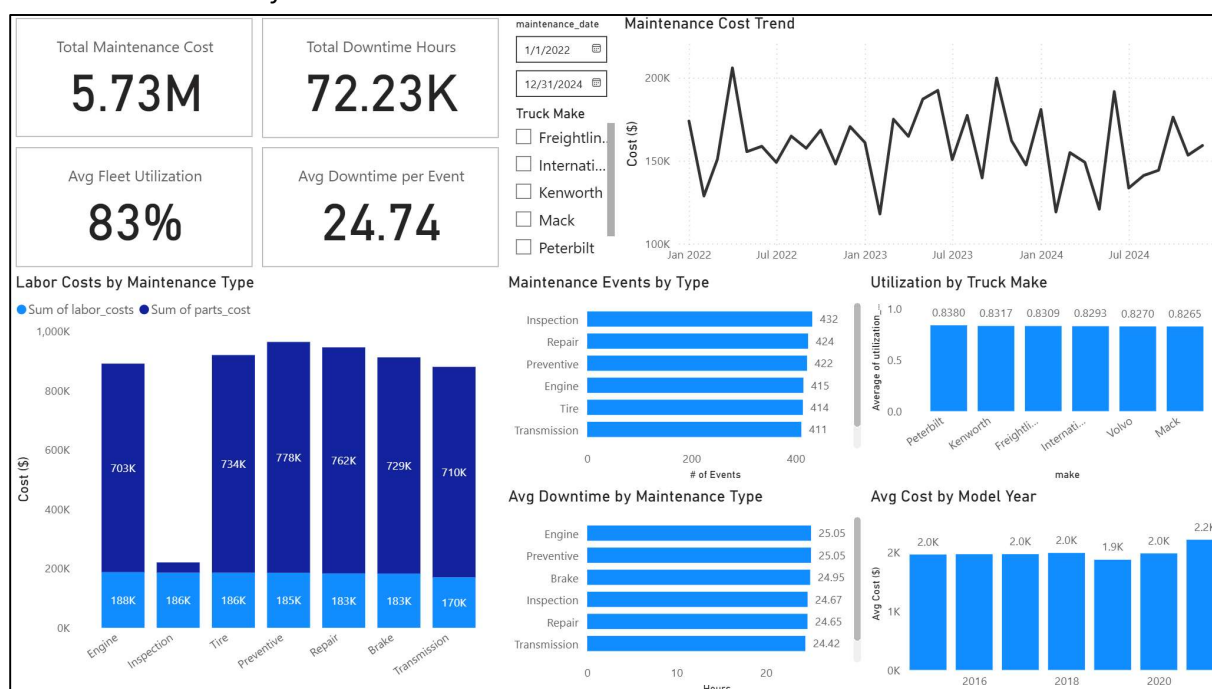
Because approximately 38% of incidents were preventable, STS8 should invest in additional driver safety training, compliance monitoring, and preventative risk management programs. Reducing preventable incidents could lower claims costs, decrease operational disruptions, and improve overall safety performance.

Fleet Maintenance Dashboard

The Fleet Maintenance Dashboard (Figure 6) was developed to provide maintenance managers and fleet supervisors with visibility into vehicle maintenance performance, fleet utilization, maintenance costs, and equipment downtime. The dashboard utilizes data from the maintenance records, vehicle, and fleet operations datasets to monitor maintenance activities and identify opportunities to improve equipment reliability and operational efficiency. By consolidating maintenance costs, downtime metrics, utilization rates, and maintenance event data into a single view, the dashboard supports proactive fleet management and informed maintenance planning.

Figure 6

Fleet Maintenance Performance Dashboard



Note. This dashboard summarizes fleet maintenance operations, including maintenance costs, downtime, fleet utilization, maintenance activity trends, event frequency, and maintenance performance by truck make, model year, and maintenance category.

Maintenance managers can use this dashboard to monitor maintenance spending, identify maintenance categories generating the highest costs, evaluate downtime trends, and assess overall fleet utilization. The dashboard also supports resource allocation decisions by highlighting maintenance event

frequency and average downtime associated with different maintenance activities. Leadership can use these insights to evaluate the effectiveness of maintenance programs, prioritize preventive maintenance efforts, and identify opportunities to reduce vehicle downtime while maintaining fleet reliability.

The analysis revealed that STS8 incurred approximately \$5.73 million in maintenance costs while experiencing more than 72,000 hours of fleet downtime during the reporting period. Despite these maintenance requirements, average fleet utilization remained relatively strong at approximately 83%, indicating that vehicles were generally available to support operational demands. Maintenance event analysis showed that inspections, repairs, preventive maintenance, and engine-related work represented the most common maintenance activities. Cost analysis further indicated that parts costs consistently represented a larger portion of maintenance expenditures than labor costs across most maintenance categories. Average downtime per maintenance event remained relatively consistent across maintenance types at approximately 24 to 25 hours per event, suggesting that vehicle availability is affected similarly regardless of maintenance category. Several important patterns emerged from the maintenance analysis. Maintenance costs remained relatively stable over time, indicating consistent maintenance requirements throughout the reporting period. Fleet utilization also remained highly consistent across truck manufacturers, suggesting that no individual vehicle brand significantly outperformed or underperformed the rest of the fleet. One notable finding is that inspection-related maintenance activities generated downtime levels comparable to those associated with major repairs and engine maintenance. This is somewhat unexpected because inspections are generally considered routine activities that should require less downtime than significant repair work. This pattern may indicate scheduling inefficiencies, delays in parts availability, administrative bottlenecks, or opportunities to streamline inspection procedures. Additionally, the relatively small variation in downtime across maintenance categories suggests that broader process improvements may be more impactful than focusing on any single maintenance type.

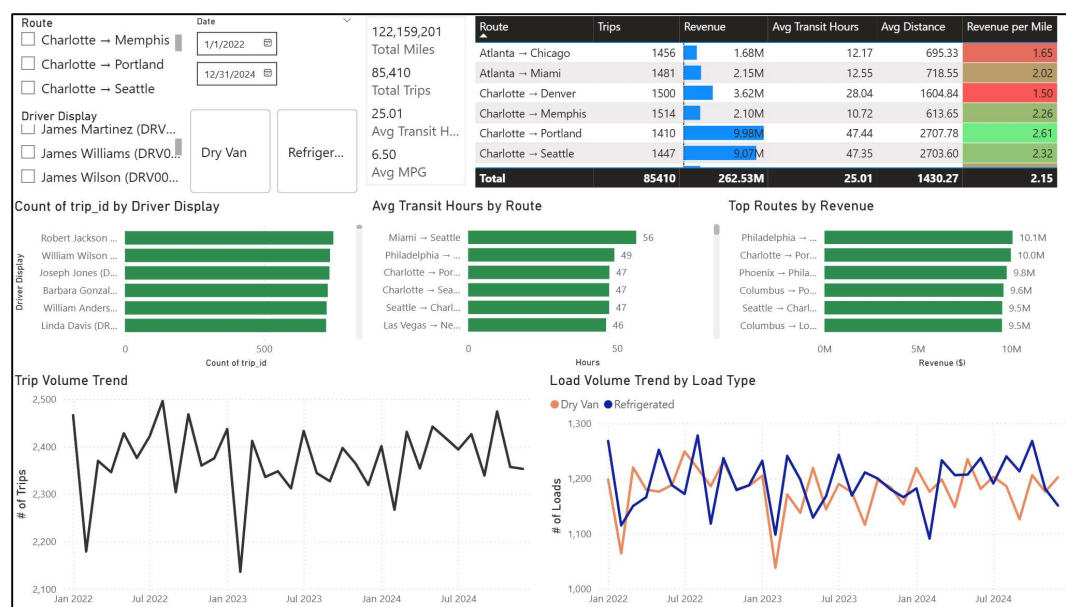
The organization should review maintenance and inspection scheduling practices to identify opportunities for reducing downtime. Predictive maintenance techniques, improved parts inventory management, and better coordination between maintenance facilities and operations teams may help reduce service interruptions while maintaining regulatory compliance.

Operations Dashboard

The Operations Dashboard (Figure 7) was developed to provide operations personnel with visibility into shipment activity, route performance, driver productivity, transportation efficiency, and freight movement throughout the STS8 Logistics network. The dashboard utilizes information from the loads, trips, routes, drivers, and truck utilization datasets to monitor operational performance and support day-to-day decision-making. Interactive filters allow users to analyze performance by route, driver, load type, and reporting period, enabling managers to identify operational issues and evaluate transportation efficiency across the network.

Figure 7

Operations Dashboard



Note. This dashboard provides operational visibility into transportation performance by summarizing fleet activity, route efficiency, driver productivity, and shipment trends. Key metrics, route-level performance indicators, and load volume patterns support operational planning and resource allocation decisions.

Operations managers can use this dashboard to monitor shipment volumes, evaluate route performance, assess driver productivity, and identify opportunities to improve transportation efficiency. Route-level metrics such as revenue per mile, average transit hours, and average route distance provide insight into operational effectiveness and profitability. Driver productivity metrics can be used to evaluate workload distribution and identify top-performing personnel. Trend visualizations allow managers to monitor freight demand over time, while load type analysis helps support capacity planning and resource allocation decisions. The dashboard can also assist managers in identifying routes that generate strong financial performance while maintaining acceptable transit times.

Analysis of the dashboard indicates that STS8 Logistics completed approximately 85,000 trips and transported more than 122 million miles of freight during the reporting period. Average transit time was approximately 25 hours per shipment, while average revenue generated approximately \$2.15 per mile traveled. Several routes generated substantially higher revenue than others, with the highest-performing routes producing nearly \$10 million in revenue during the reporting period. Shipment volume remained relatively stable throughout the analysis period, averaging between 2,200 and 2,500 loads per month. Route-level analysis further revealed meaningful differences in transit times, route distances, and revenue generation, demonstrating that operational performance varies significantly across the transportation network. Several important patterns emerge from the operational analysis. Shipment volumes remained relatively consistent throughout the reporting period, suggesting stable customer demand and transportation activity. The load volume trend by freight type indicates that refrigerated and dry van shipments followed similar patterns over time, reflecting balanced demand across both service offerings. Route-level analysis reveals notable differences in revenue per mile, with some routes generating substantially greater financial returns than others despite similar operating

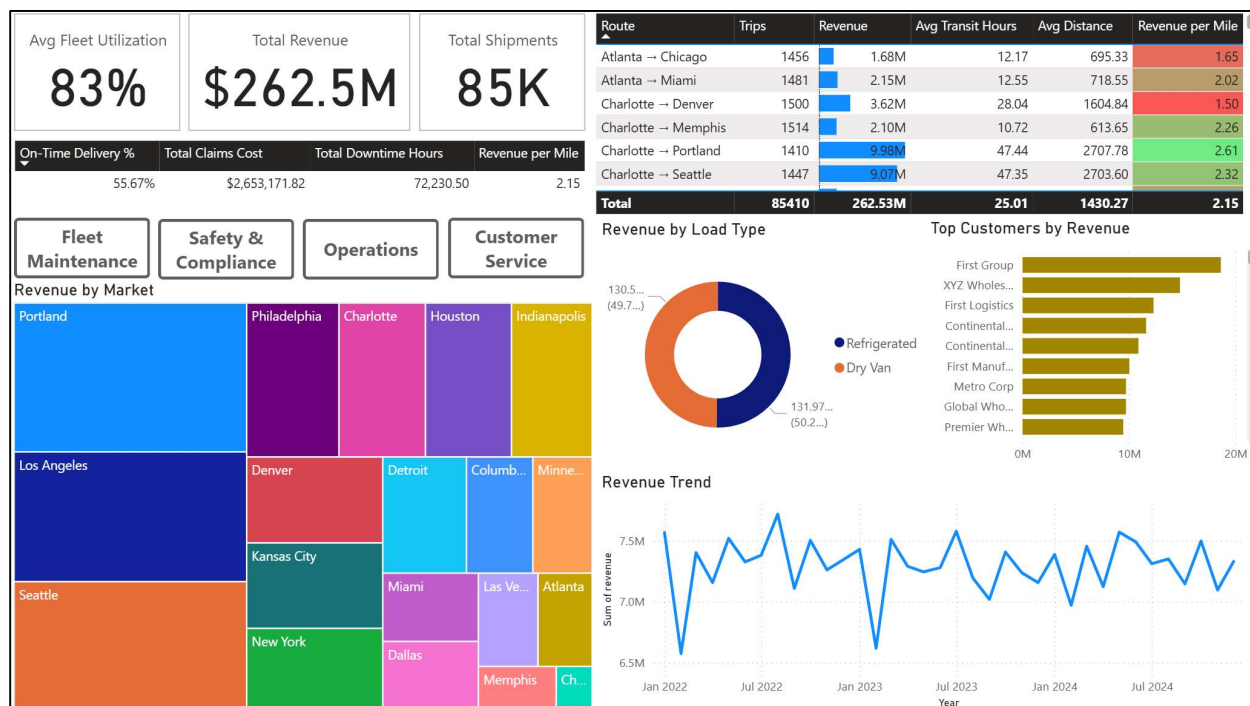
characteristics. Average transit times also varied considerably across routes, highlighting potential opportunities to improve scheduling efficiency and route planning. One notable pattern is the concentration of revenue among a relatively small number of routes. Several routes consistently generated significantly higher revenue than the remainder of the transportation network, suggesting that these lanes represent strategically important corridors for the organization. Additionally, the route performance matrix indicates that longer routes do not always produce proportionally higher revenue, suggesting opportunities to evaluate route profitability and optimize network operations. These findings provide management with actionable insights that can support transportation planning, capacity management, and operational improvement initiatives.

STS8 should continue developing relationships with new customers and expanding into additional geographic markets. Diversifying the customer portfolio would reduce dependence on a limited number of high-value accounts and improve long-term financial stability.

Executive Dashboard

The Executive Dashboard (Figure 8) was developed to provide senior leadership with a high-level view of organizational performance across customer service, operations, safety, and fleet management functions. Unlike the stakeholder-specific dashboards, this dashboard consolidates key performance indicators and summarizes critical business metrics in a single location to support strategic decision-making. The dashboard utilizes data from shipment, customer, route, safety, and maintenance datasets and incorporates dashboard navigation buttons that allow executives to drill into the Customer Service, Safety & Compliance, Operations, and Fleet Maintenance dashboards for additional detail when needed.

Figure 8
Executive Performance Dashboard



Note. This dashboard provides a high-level view of organizational performance by integrating key financial, operational, customer service, safety, and fleet management metrics. Executives can monitor overall business health, evaluate revenue performance, identify high-value markets and customers, assess route profitability, and navigate to detailed dashboards for further analysis.

Executives can use this dashboard to monitor overall organizational health and quickly identify areas that may require additional attention. Key metrics such as total revenue, shipment volume, fleet utilization, claims costs, downtime hours, revenue per mile, and on-time delivery performance provide a balanced view of financial, operational, and service outcomes. The dashboard serves as an executive summary that highlights performance trends while providing direct access to more detailed stakeholder dashboards. This approach allows leadership to move from high-level monitoring to detailed analysis without becoming overwhelmed by operational details.

The analysis indicates that STS8 Logistics generated approximately \$262.5 million in revenue while completing more than 85,000 shipments and maintaining an average fleet utilization rate of 83%. Revenue per mile averaged approximately \$2.15, providing insight into transportation profitability across the network. On-time delivery performance averaged 55.67%, while total claims costs exceeded \$2.6 million and fleet downtime exceeded 72,000 hours during the reporting period. Revenue analysis shows that a relatively small number of customers and markets account for a substantial portion of total revenue. Similarly, route-level performance metrics demonstrate meaningful differences in revenue generation and operational efficiency across the transportation network.

Several important trends emerge from the executive-level analysis. Revenue remained relatively stable throughout the reporting period, suggesting consistent customer demand and business activity. Revenue distribution by load type was nearly balanced between refrigerated and dry van freight, indicating a diversified service portfolio rather than dependence on a single freight segment. Geographic revenue analysis shows that certain markets, including Portland, Los Angeles, Seattle, and Philadelphia, contribute significantly more revenue than other locations, highlighting strategically important regions for the organization. The dashboard also reveals several areas that may warrant further investigation. While revenue and fleet utilization remain strong, on-time delivery performance appears substantially lower than desired, suggesting potential operational or scheduling challenges. Additionally, the combination of significant claims costs and downtime hours may indicate opportunities for safety improvements and maintenance optimization. Because these metrics are linked directly to the supporting stakeholder dashboards, executives can quickly investigate underlying causes by navigating to the relevant Customer Service, Safety & Compliance, Operations, or Fleet Maintenance dashboard. The use of familiar visualizations and consistent KPI measures across dashboards further supports efficient analysis and promotes alignment between operational managers and executive leadership.

The executive dashboard should be adopted as a recurring performance management tool. Leadership should review key performance indicators including revenue per mile, fleet utilization, on-time delivery percentage, claims costs, and downtime hours on a regular basis. Monitoring these metrics over time will help leadership identify emerging issues early and make more informed strategic decisions.

While the dashboards presented in this project primarily support descriptive and diagnostic reporting, STS8 Logistics should continue expanding its reporting capabilities as governance practices mature and additional historical data becomes available. Predictive analytics could be incorporated to forecast shipment delays, maintenance requirements, customer demand, and other operational trends before they occur. These forecasts would allow managers to proactively allocate resources and address potential issues before they affect organizational performance. STS8 could also implement exception-based alerts that automatically notify stakeholders when predefined performance thresholds are exceeded, such as declining on-time delivery rates, increasing maintenance costs, or elevated safety incidents. These capabilities would complement the existing reporting environment by supporting more proactive decision-making and continuous operational improvement.

Implementation Plan

Implementation is one of the most costly and time-consuming phases in the systems development life cycle (SDLC) due to the number of people involved and the significant labor required to complete the project (Valacich & George, 2020). Because of this, careful planning is essential to ensure the project stays on track. An implementation plan serves as the roadmap for transitioning a proposed solution from design into day-to-day operations. It establishes the project phases, timelines, milestones, deliverables, stakeholder responsibilities, and supporting activities needed to deploy the solution successfully while minimizing operational disruption. Because implementation involves coordinating technical resources, business stakeholders, organizational processes, and user adoption, it is often one of the most resource-intensive phases of a systems project.

For STS8, a well-developed implementation plan is essential to ensure the proposed business intelligence environment is deployed in a structured and controlled manner. While the previous sections of this project focused on designing the organization's data governance framework, reporting environment, and Power BI dashboards, these components provide value only if they are successfully integrated into daily operations. A structured implementation plan helps coordinate project activities, prepare stakeholders for the transition, and establish the foundation for long-term adoption.

Project Plan

The project plan establishes the framework for implementing STS8's proposed business intelligence solution. It provides a structured approach for moving the project from planning into production while maintaining alignment with the organization's business objectives. Breaking the implementation into defined phases also allows progress to be monitored throughout the project. As each phase is completed, the project team can confirm that key objectives have been achieved before moving to the next stage. This approach reduces uncertainty during implementation and provides a clear path toward deployment.

Project Management Methodology

Selecting an appropriate project management methodology is an important part of the implementation process. Different methodologies are designed to address different types of projects,

and no single approach is appropriate for every implementation. The proposed business intelligence environment at STS8 includes both technical infrastructure and end-user reporting tools. As a result, the implementation requires a methodology that provides structure during foundational activities while remaining flexible during dashboard development and user testing.

The Waterfall methodology follows a sequential approach in which each phase of the project is completed before the next phase begins. Requirements are established early in the project and major deliverables are approved before additional work proceeds (Stair & Reynolds, 2018). Because each phase builds upon the previous one, Waterfall works well when project requirements are stable and changes are expected to be minimal. This approach also provides clear documentation and defined milestones throughout the implementation process. Agile takes a different approach by emphasizing iterative development and continuous collaboration with stakeholders. Rather than completing the entire solution before obtaining feedback, Agile delivers smaller portions of the project over multiple development cycles. Scrum is one of the most common Agile frameworks and organizes work into short, iterative sprints that produce functional increments of the system (Valacich & George, 2020). Frequent stakeholder involvement allows the development team to evaluate progress, respond to changing business needs, and continuously improve the solution throughout implementation. Scrum also establishes defined team roles that promote collaboration, transparency, and continuous improvement during the project lifecycle (Stair & Reynolds, 2018).

Neither methodology independently provides the best fit for the proposed STS8 implementation. Portions of this project require a structured, sequential approach. Data governance policies should be finalized before implementation begins. Likewise, the data warehouse, data integration processes, and security controls should be established before reports are developed. Allowing these elements to change throughout development would increase project risk and could affect the reliability of the reporting environment. Other portions of the implementation benefit from a more iterative process. Dashboard development is rarely complete after the first design because reporting needs often change once users begin interacting with the data. Executive leadership may request additional strategic metrics. Operations managers may identify opportunities to improve visualizations or modify existing KPIs. Customer service personnel may request different reporting views that better support daily operations. An iterative development process allows these changes to be incorporated before the solution is deployed across the organization.

For these reasons, STS8 will adopt a hybrid project management methodology. Rather than applying a single framework across the entire implementation, the project will use the methodology that best aligns with the work being performed. Project initiation, governance development, data warehouse implementation, and system configuration will follow a predictive Waterfall approach because these activities require clear requirements, formal approvals, and stable deliverables. Dashboard development, report refinement, and user acceptance testing will follow Scrum so stakeholder feedback can be incorporated throughout development. According to the Archer & Kaufman (2013), hybrid project management is a fit-for-purpose approach that combines predictive and Agile practices based on the specific objectives, risks, and characteristics of a project rather than relying on a single methodology for every phase. That philosophy aligns well with STS8's implementation because it provides the structure

needed to establish a reliable business intelligence foundation while maintaining the flexibility needed to refine reporting solutions as organizational needs evolve.

Project Phases

Implementing a business intelligence solution is not a single event. It is a structured process that builds the solution one step at a time. Organizing the project into defined phases allows STS8 to focus on one objective before moving to the next. It also provides opportunities to evaluate progress throughout the implementation. If issues are identified, they can be addressed before they affect later stages of the project.

Table 3 presents the six phases that will guide the implementation of the proposed solution. Each phase has a specific purpose and concludes with measurable success criteria. Completion of one phase establishes the foundation for the next phase to begin. This approach helps maintain project control throughout the implementation while reducing unnecessary risk. It also creates clear accountability by defining the responsibilities of the stakeholders involved in each stage of the project.

Table 3

Project Phases for the STS8 Implementation Plan

Project Phase	Purpose	Primary Stakeholders	Major Activities	Dependencies	Success Criteria
Project Initiation	Establish the project's scope, objectives, and governance structure before implementation begins.	Executive Sponsor, Project Manager, Executive Leadership	Develop the project charter, identify stakeholders, assign project roles, secure executive approval, and allocate project resources.	Approved business case and executive sponsorship.	Project charter approved, project team established, and project funding authorized.
Planning & Design	Validate business requirements and finalize the technical design for the business intelligence environment.	Project Manager, Data Governance Committee, IT, Business Stakeholders	Confirm reporting requirements, finalize data governance policies, review solution architecture, complete risk assessment, and develop the implementation schedule.	Project initiation completed and project charter approved.	Requirements approved, governance framework finalized, and implementation plan accepted.
Development & Configuration	Build and configure the technical components that support the business intelligence solution.	BI Developers, Data Engineers, IT Support, Data Stewards	Develop the data warehouse, integrate data sources, configure Power BI, establish role-based security, and develop initial dashboards.	Planning and design completed with approved technical specifications.	Data warehouse operational, integrations validated, dashboards developed, and security configuration verified.
Testing	Validate system functionality, data accuracy, and user readiness before deployment.	BI Team, IT Support, Business Stakeholders	Perform system testing, validate data quality, conduct user acceptance testing (UAT), evaluate performance, and resolve identified issues.	Development and configuration completed.	User acceptance testing completed, critical defects resolved, and stakeholder approval received for production deployment.

Deployment	Transition the solution into the production environment while minimizing disruption to daily operations.	Project Manager, IT Support, Business Units	Deploy the production environment, migrate users, configure production access, monitor system performance, and provide initial implementation support.	Testing completed and production approval granted.	Production environment operational, users successfully transitioned, and system performance meets established requirements.
Post-Implementation Review	Evaluate project outcomes and establish a plan for continuous improvement.	Executive Leadership, Project Manager, Data Governance Committee	Review project objectives, document lessons learned, evaluate project performance, collect stakeholder feedback, and identify future enhancement opportunities.	Successful production deployment and operational stabilization.	Project objectives achieved, lessons learned documented, and continuous improvement roadmap approved.

Note. This table outlines the six project phases that guide the implementation of STS8's proposed business intelligence solution. Each phase identifies its primary purpose, major activities, dependencies, responsible stakeholders, and success criteria. Together, these phases establish a structured framework for managing the implementation from project initiation through post-implementation review.

Table 3 presents the six phases that will guide the implementation of the proposed solution. Each phase has a specific purpose and concludes with measurable success criteria. Completion of one phase establishes the foundation for the next phase to begin. This approach helps maintain project control throughout the implementation while reducing unnecessary risk. It also creates clear accountability by defining the responsibilities of the stakeholders involved in each stage of the project.

Project Milestones & Deliverables

Project milestones (Table 4), serve as measurable checkpoints that indicate progress throughout the implementation. Unlike project phases, which consist of a series of activities completed over time, milestones represent significant achievements that demonstrate the project is ready to advance to the next stage. Establishing these checkpoints helps STS8 monitor implementation progress, verify that project objectives are being met, and confirm that key deliverables have been completed before additional work begins. Each milestone represents a significant point in the implementation where project progress is evaluated before moving forward. This approach provides opportunities to identify potential issues early and confirm that project objectives continue to align with organizational expectations. Requiring formal approval at each milestone also strengthens accountability by ensuring that technical and business stakeholders remain engaged throughout the implementation process.

Table 4
Major Project Milestones for STS8 Implementation Plan

Milestone	Associated Phase	Primary Deliverables	Approving Stakeholder	Completion Criteria
Project Charter Approved	Project Initiation	Approved Project Charter	Executive Sponsor	Project formally authorized
Governance Framework Approved	Planning & Design	Governance Framework	Data Governance Committee	Governance policies adopted

Data Warehouse Completed	Development & Configuration	Operational Data Warehouse	IT Director	All data sources successfully integrated
Power BI Dashboard Development Completed	Development & Configuration	Dashboard Suite	BI Development Team	Dashboards meet functional requirements
User Acceptance Testing (UAT) Approved	Testing	UAT Report	Business Stakeholders	Users approve the solution for deployment
Product Go-Live	Deployment	Production Business Intelligence Environment	Project Manager	Solution successfully deployed
Post-Implementation Review Completed	Post-Implementation Review	Lessons Learned Report	Executive Sponsor	Project formally closed and future improvements identified

Note. This table identifies the major milestones used to monitor progress throughout the implementation of STS8's proposed business intelligence solution. Each milestone represents a significant project checkpoint that confirms key deliverables have been completed, stakeholder approval has been obtained, and the project is ready to advance to the next phase of implementation.

Reaching project milestones demonstrates that implementation activities have been completed, but successful implementation should also be evaluated after the solution is deployed. STS8 will establish measurable post-implementation indicators that evaluate both system adoption and organizational effectiveness. Examples include dashboard adoption rates, user satisfaction survey results, report utilization, help desk ticket volume, and the time required for managers to access information needed for operational decision-making. Monitoring these measures after deployment allows leadership to determine whether the business intelligence environment is delivering the expected organizational value while identifying opportunities for additional training, system enhancements, or process improvements.

Implementation Timeline

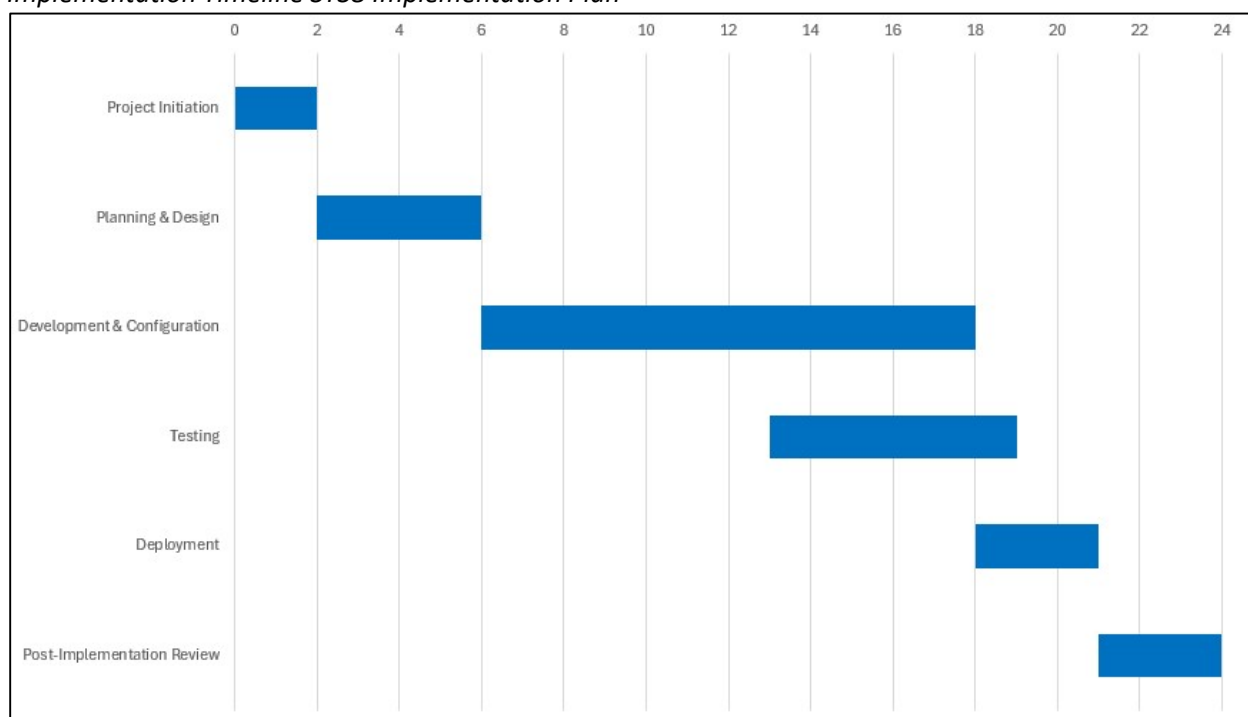
The implementation timeline establishes a realistic schedule for deploying STS8's proposed business intelligence solution while maintaining alignment with the project's hybrid management methodology. As shown in Figure 9, the implementation is expected to span approximately 24 weeks. This schedule provides sufficient time to complete project planning, develop the technical infrastructure, conduct iterative testing, and prepare the organization for deployment. It also allows project resources and stakeholder involvement to be coordinated throughout the implementation rather than concentrated at the end of the project.

Figure 9 illustrates how the implementation transitions from a structured planning approach to a more iterative development process. The early phases focus on project initiation, planning, and establishing the technical foundation needed to support the business intelligence environment. As

development progresses, testing begins before all development activities have concluded. This overlap allows completed dashboards and system components to be validated while additional features continue to be developed and refined. Deployment activities also begin before testing has completely concluded, reflecting a phased rollout that prepares the production environment while final validation activities are completed. Organizing the schedule in this manner supports the hybrid project management methodology by balancing structured planning with continuous stakeholder feedback and iterative improvement throughout implementation.

Figure 9

Implementation Timeline STS8 Implementation Plan



Note. This figure illustrates the planned implementation timeline for STS8's proposed business intelligence solution. The schedule identifies the sequence and estimated duration of each project phase over a 24-week implementation period.

Risk Management

Risk management is an essential component of any implementation plan because it helps organizations identify potential issues before they significantly affect project objectives. As Dionisio (2021) explains, risk management is one of the most challenging aspects of project management, but it is also one of the most important because it enables project teams to anticipate problems and develop strategies to reduce their impact. Rather than reacting to issues after they occur, STS8 should continuously evaluate project risks throughout the implementation lifecycle and apply mitigation strategies that minimize disruptions to project cost, schedule, and quality.

Table 3 identifies the primary risks associated with implementing STS8's proposed business intelligence solution and the strategies recommended to reduce their potential impact. Data quality presents one of the most significant technical risks because inaccurate or inconsistent information can reduce confidence in dashboards and lead to poor business decisions. To address this risk, organizational

data should be validated and cleansed before migration into the data warehouse, and ongoing monitoring should continue after deployment to maintain reporting accuracy. Integration challenges also represent a potential risk because the proposed solution depends on consolidating information from multiple operational systems. Phased integration testing and early validation activities can help identify compatibility issues before they affect production operations.

Successful implementation also depends on organizational readiness. Resistance to change is common when employees are asked to adopt new technologies or modify established business processes. Likewise, inadequate training can reduce user confidence and limit adoption of the new reporting environment. Stair and Reynolds (2018) note that user acceptance is one of the most significant challenges associated with implementing new information systems because employees are often reluctant to change familiar processes. STS8 can reduce these risks by communicating the benefits of the new business intelligence environment early, involving stakeholders throughout development, and providing role-specific training before deployment. Project risks should be reviewed regularly throughout implementation so that emerging issues can be identified, mitigation strategies can be adjusted, and corrective actions can be taken before they affect project success.

Table 5
Risk Management Plan STS8 Implementation Plan

Risk	Potential Impact	Likelihood	Mitigation Strategy	Responsible Stakeholder
Data quality issues	Inaccurate reporting	Medium	Conduct data profiling, validation, and cleansing before migration	Data Engineering Team
User resistance	Low system adoption	High	Implement change management plan, stakeholder communication, and training	Project Manager
Integration failures	Project delays	Medium	Perform phased integration testing and contingency planning	IT Support
Scope creep	Increased cost and schedule delays	Medium	Formal change control process and stakeholder approval	Project Manager
Security vulnerabilities	Unauthorized data access	Low	Role-based security, access reviews, and security testing	IT Security
Resource constraints	Missed deadlines	Medium	Secure executive sponsorship, prioritize project activities, and monitor resource utilization throughout implementation	Executive Sponsor

Note. This table identifies the primary risks associated with implementing STS8's proposed business intelligence solution. Each risk includes its potential impact, likelihood of occurrence, mitigation strategy, and the stakeholder responsible for monitoring and managing the risk throughout the implementation lifecycle.

Change Management

Implementing a business intelligence solution requires more than deploying new technology. It also requires preparing employees to adopt new processes, reporting practices, and decision-making tools. Change management provides a structured approach for guiding both individuals and the organization through that transition. Rather than focusing solely on the technical aspects of implementation, change management emphasizes the human side of organizational change. It helps employees understand why the change is necessary, prepares them to work within the new environment, and provides the support needed to build confidence in new systems and processes. Phillips and Klein (2023) explain that effective change management incorporates leadership support, stakeholder engagement, communication, training, and reinforcement to improve the likelihood of successful organizational change. These activities help create organizational alignment and encourage employees to adopt new ways of working instead of returning to familiar processes. Without a structured change management strategy, organizations may experience resistance, inconsistent system adoption, and an inability to realize the full benefits of a new technology investment. For STS8, successful implementation depends not only on deploying the proposed business intelligence environment but also on ensuring that employees trust the information being provided and incorporate the new reporting tools into their daily decision-making.

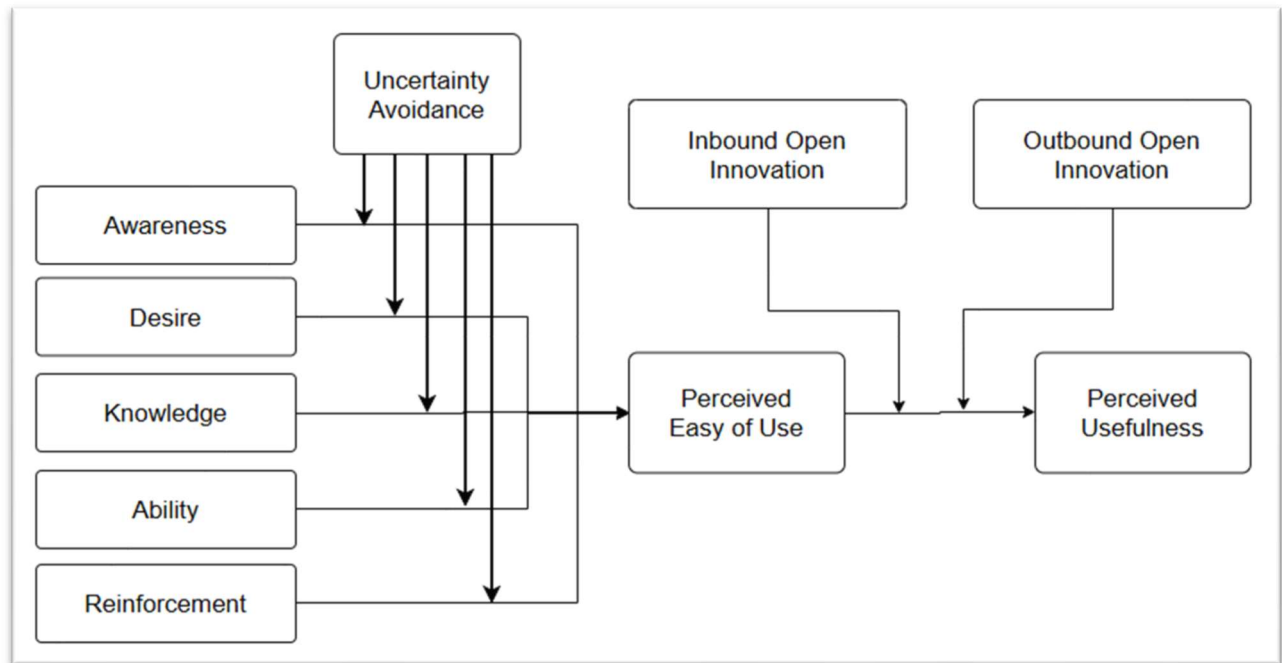
Change Management Model

Several well-established models have been developed to guide organizational change, each emphasizing different aspects of the transition process. Kotter's Eight-Step Change Model focuses on creating organizational momentum by establishing a sense of urgency, building a guiding coalition, communicating a shared vision, and reinforcing change until it becomes part of the organizational culture (Kotter, 2012). The model is particularly effective for large-scale organizational transformations because it emphasizes leadership, strategic alignment, and long-term cultural change. Lewin's Three-Stage Model takes a broader organizational perspective by describing change as a process of unfreezing existing behaviors, implementing new practices, and refreezing those behaviors to establish a new organizational norm (Lewin, 1947). Although both models provide valuable guidance for organizational change, neither directly addresses how individual employees progress through adopting new technologies or business processes.

The implementation of STS8's business intelligence environment depends heavily on individual user adoption. Employees must learn to access new dashboards, interpret business intelligence reports, and incorporate data-driven decision-making into their daily responsibilities. For this reason, the ADKAR Model (Figure 10) provides a better fit for the proposed implementation. Rather than concentrating primarily on organizational transformation, ADKAR focuses on the individual employee and the factors that influence successful adoption of change. Ali et al. (2021) found that the model provides a structured, people-focused approach that improves technology acceptance by guiding employees through each stage of the transition process. This emphasis aligns closely with STS8's implementation because the long-term success of the business intelligence environment depends on employees consistently using the new reporting tools and trusting the information they provide.

Figure 10

ADKAR Conceptual Framework



Note. Adapted from “The power of ADKAR change model in innovative technology acceptance under the moderating effect of culture and open innovation” by M. A. Ali, A. Mahmood, U. Zafar, and M. Nazim, 2021, *LogForum*, 17(4), p. 491. <https://doi.org/10.17270/J.LOG.2021.623>

As shown in Table 4, ADKAR consists of five sequential components: Awareness, Desire, Knowledge, Ability, and Reinforcement (Ali et al., 2021). Awareness ensures that employees understand why the organization is implementing the new business intelligence environment and how it supports broader organizational objectives. Desire focuses on building employee commitment by demonstrating how the solution will improve daily work rather than simply introducing another technology platform. Knowledge provides employees with the information and training necessary to perform their responsibilities within the new environment. Ability emphasizes applying those skills successfully through practice, coaching, and ongoing support during implementation. Finally, Reinforcement helps ensure that adoption continues after deployment through leadership support, performance measurement, recognition, and continuous feedback. Together, these five components provide STS8 with a structured framework for managing the human side of implementation while encouraging long-term adoption of the proposed business intelligence solution.

Table 6

ADKAR Framework for STS8 Change Management

ADKAR Elements	Application to Power BI Implementation
Awareness	• Awareness of the need for change. • Understanding why Power BI is being implemented.
Desire	• Desire to participate in and support the change. • Motivating staff to embrace Power BI.
Knowledge	• Knowledge about how to change.

	• Training users on how to use Power BI effectively.
Ability	• Ability to implement required skills and behaviors. • Using Power BI in daily workflows.
Reinforcement	• Reinforcement to sustain the change. • Ongoing support, recognition, or performance incentives

Note. This table illustrates how the ADKAR change management model can be applied to support the implementation for STS8.

Stakeholder Engagement & Executive Sponsorship

Stakeholder engagement strengthens organizational change by encouraging participation throughout the implementation process rather than treating employees as passive recipients of change (Phillips & Klein, 2023). Building that sense of ownership early in the project helps create organizational buy-in and increases the likelihood that employees will adopt the new business intelligence solution after implementation. Because of this, stakeholder engagement should begin during the earliest stages of the implementation rather than after major project decisions have already been made. Involving stakeholders throughout the project gives them an opportunity to contribute to the design of the new business intelligence environment and helps ensure that the solution supports day-to-day business operations. For STS8, key stakeholders include executive leadership, department managers, IT personnel, and the employees who will use the dashboards and reports each day. These groups should participate during requirements validation, dashboard reviews, and user acceptance testing so their feedback can be incorporated before the solution is deployed. This collaborative approach helps identify potential issues early while giving employees a sense of ownership over the new reporting environment.

On the other hand, executive sponsorship provides the leadership needed to sustain stakeholder engagement throughout the project. Senior leadership should clearly communicate why STS8 is investing in a business intelligence environment and explain how the solution supports the organization's strategic goals. Employees are more likely to support organizational change when they understand the purpose of the project and recognize how it will improve decision-making within their own departments. Executive sponsors should reinforce these messages throughout implementation by participating in project updates, demonstrating visible support for the initiative, and encouraging managers to discuss project progress with their teams. Department managers play an equally important role because they translate organizational objectives into daily operations and help employees understand how the new reporting tools will improve existing business processes. The success of organizational change depends on leaders who consistently communicate the vision for change and maintain organizational commitment throughout the transition (Kotter, 2012).

Communication Strategy

A structured communication strategy helps ensure that stakeholders remain informed and engaged throughout the implementation process. Rather than viewing communication as a series of project updates, STS8 should establish a communication plan that identifies what information will be shared, who will receive it, and when communication should occur. Archer & Kaufman (2013) recommend engaging stakeholders early and maintaining communication throughout the project lifecycle so expectations remain aligned and stakeholders have opportunities to provide feedback as implementation progresses. A consistent communication strategy also reduces uncertainty by helping

employees understand the purpose of the project, the progress being made, and the changes they should expect as the business intelligence environment is introduced.

STS8 should begin communication during the Project Initiation phase with a cross-functional kickoff meeting involving executive leadership, department managers, IT personnel, and key end users. This meeting should introduce the business intelligence initiative, explain its objectives, and outline the overall implementation timeline. As the project moves through development, communication should continue through regular project updates and department meetings that summarize completed work, upcoming milestones, and any activities that may affect business operations. Because STS8 is using a hybrid implementation methodology, communication should also occur at the conclusion of each development iteration. Demonstrating completed dashboards and reporting capabilities allows stakeholders to evaluate project progress while validating that the solution continues to meet business requirements. These demonstrations also create opportunities for employees to ask questions and provide feedback before additional functionality is developed. By integrating communication activities into each phase of the implementation, STS8 can maintain stakeholder engagement while reducing uncertainty throughout the transition.

Managing Resistance

Resistance to change is one of the most significant risks associated with implementing a new business intelligence environment because success depends on employees adopting new processes rather than returning to familiar reporting methods (Stair & Reynolds, 2018). Employees may be reluctant to replace established workflows if they do not understand the purpose of the change or believe the new system provides little value to their daily responsibilities. Resistance is often influenced by cultural factors, unclear performance metrics, and the perception that business intelligence tools provide limited individual or organizational benefit (BARC, 2017). Similar challenges were identified during a previous systems implementation analysis, where resistance developed because employees viewed new processes as disruptive to established routines rather than improvements to existing operations. These findings demonstrate that managing resistance requires more than implementing new technology. It also requires helping employees understand why the change is necessary and how it supports organizational success.

Identifying resistance early allows STS8 to address employee concerns before they become barriers to implementation. Readiness assessments should be conducted during the planning phase to evaluate employee expectations and identify departments that may require additional support. As the project progresses, feedback gathered during dashboard demonstrations, sprint reviews, and user acceptance testing should be used to identify recurring concerns and determine whether modifications are needed before deployment. Department managers should also encourage open discussions during team meetings so employees have opportunities to raise questions about new reporting processes or dashboard functionality. Monitoring employee feedback throughout the implementation allows the project team to identify patterns of resistance while maintaining communication with those most affected by the change (Valacich & George, 2020). Once potential resistance has been identified, STS8 should respond through targeted communication, active stakeholder involvement, and role-specific support. Effective change management helps individuals adapt their behaviors, expectations, and actions throughout the transition rather than expecting acceptance to occur automatically (Wick, 2022). Employees who express concerns should be encouraged to participate in dashboard reviews and validation activities so they can contribute directly to improving the solution before deployment. Additional coaching and focused training should also be provided when employees demonstrate uncertainty using the new reporting tools. This collaborative approach encourages employees to become

active participants in the implementation and reinforces the stakeholder engagement and communication strategies discussed earlier.

Successfully managing resistance will improve long-term adoption of the proposed business intelligence environment. Employees who understand the value of the new reporting tools and feel confident using them are more likely to incorporate data into their daily decision-making. Greater user adoption also promotes consistent reporting across the organization because departments rely on standardized dashboards instead of independently maintained spreadsheets or manually generated reports (Stair & Reynolds, 2018). As confidence in the system increases, STS8 will be better positioned to develop a data-driven culture that supports operational efficiency and more informed strategic decision-making.

Change Adoption & Continuous Improvement

Long-term success depends on whether employees continue using the new reporting tools and incorporate data-driven decision-making into their daily responsibilities. Maintaining user adoption requires ongoing leadership support, regular evaluation of system performance, and a commitment to continuous improvement (Stair & Reynolds, 2018). Reinforcing these behaviors after implementation helps ensure that employees do not return to previous reporting methods and that the organization realizes the full value of its investment in business intelligence. STS8 should evaluate user adoption through a combination of system usage metrics, employee feedback, and operational performance indicators. Dashboard usage statistics can help determine whether employees are actively using the new reporting environment, while user surveys and manager feedback can identify areas where additional support may be needed. The project team should also review help desk requests and enhancement recommendations to identify recurring issues that may affect adoption. Measuring these indicators provides valuable insight into how effectively the business intelligence environment is being integrated into daily operations and allows leadership to address concerns before they impact long-term success.

Continuous improvement should remain an ongoing organizational responsibility after the implementation has been completed. The Post-Implementation Review identified in the project plan provides an opportunity to evaluate project outcomes, document lessons learned, and identify enhancements that improve reporting capabilities or user experience. As business needs evolve, dashboards and performance metrics should also be reviewed to ensure they continue supporting operational and strategic decision-making. Maintaining an open feedback process encourages employees to contribute ideas for future improvements while reinforcing their role in the continued success of the business intelligence environment. This commitment to continuous improvement will help STS8 sustain user adoption, strengthen organizational confidence in its reporting environment, and continue building a data-driven culture. Collectively, stakeholder engagement, executive sponsorship, communication, resistance management, and continuous improvement provide STS8 with a comprehensive change management strategy that supports successful implementation and long-term adoption of the proposed business intelligence environment.

Training & Support

In order to have a successful implementation, employees must develop the knowledge and confidence needed to incorporate the new reporting tools into their daily responsibilities. Gupta (2024) explains that successful business intelligence implementations depend on preparing end users to understand the capabilities of the system and positioning them to effectively utilize those capabilities within their daily work. A structured training and support program provides users with the skills necessary to navigate and trust the new environment. Because employees have different technical

backgrounds and varying levels of experience with business intelligence tools, training should be tailored to the responsibilities of each user group rather than delivered through a single standardized program. STS8 will establish a Learning and Development (L&D) function responsible for planning, delivering, and maintaining the organization's training program. The L&D team will work closely with the project manager, BI developers, IT personnel, and department managers to ensure that training materials reflect both the technical features of the system and the operational needs of each department. Training activities will begin before the production rollout so employees have opportunities to become familiar with the new reporting environment before it becomes part of their daily responsibilities. Following deployment, the L&D function will continue providing refresher training, updated learning materials, and ongoing educational resources as new features are introduced or business requirements evolve. Viewing training as a continuous process rather than a one-time implementation activity will help STS8 strengthen user confidence, encourage long-term adoption, and maximize the value of its business intelligence investment.

Audience & Training Scope

Training should be tailored to the responsibilities of each user group because employees interact with the business intelligence environment in different ways. While all users require a basic understanding of the system and its reporting capabilities, the depth and focus of training should reflect each group's operational responsibilities. Aligning training with individual job responsibilities helps employees develop the skills most relevant to their daily work while improving overall user confidence and adoption.

Table 7

Role-Base Training Overview

Audience	Training Objectives	Key Topics Covered
Executive Leadership	Develop the ability to interpret organizational performance and support strategic decision-making	Executive dashboards, KPIs, trend analysis, performance monitoring
Executive Sponsor	Monitor implementation progress and champion organizational adoption	Executive dashboards, project KPIs, adoption metrics, governance reporting
Department Managers	Use BI reports to monitor operations and improve departmental performance	Department dashboards, filtering reports, exception reporting, operational metrics
Operational End Users	Access customer information and monitor shipment activity	Shipment status, customer activity dashboards, service metrics, issue reporting
BI Developers / IT	Administer and maintain the BI environment	Power BI administration, gateway management, security, troubleshooting, report deployment
Data Stewards	Ensure data quality and governance compliance	Data quality monitoring, governance policies, metadata management, data validation

Note. This table outlines the primary user groups within STS8, their specialized training focus, and the concepts most relevant to their organizational responsibilities.

Training Methods

Different training methods should be incorporated into the implementation because employees perform different job functions and require varying levels of technical instruction. More complex topics, such as system administration, dashboard development, data governance, and security management, are best delivered through instructor-led workshops and hands-on training environments where employees can ask questions, practice real-world tasks, and receive immediate feedback from subject matter experts. Less complex topics, including dashboard navigation, report filtering, and common user tasks, can be reinforced through self-paced learning resources such as video tutorials and quick-reference guides. These resources allow employees to review material at their own pace and revisit concepts after formal training has been completed. Combining multiple delivery methods creates a more flexible learning environment while accommodating different learning styles, technical experience levels, and work schedules. It also ensures that employees continue to have access to training resources after implementation, reducing reliance on technical support for routine questions and helping reinforce long-term user adoption.

Table 8
Training Methods and Resources Required

Training Method	Description	Resources Required
Instructor-Led Workshops	Interactive sessions introducing the business intelligence environment and demonstrating core functionality.	Training room or Teams meeting, presentation materials, sample dashboards
Hands-On Training Environment	Sandbox environment allowing employees to practice using dashboards and reports with sample data.	Power BI training workspace, sample datasets, demo accounts
Video Tutorials	Short, task-focused videos covering common reporting activities and dashboard navigation.	Learning Management System (LMS) or shared drive, video hosting platform, audio device
Quick-Reference Guides	Printable and digital job aids summarizing common tasks and troubleshooting tips.	PDFs and/or printable handouts, SharePoint for ongoing updates

Note. This table identifies the primary training methods and supporting resources that will be used to prepare STS8 employees for the implementation of the proposed business intelligence environment. Using multiple delivery methods provides users with both structured instruction and ongoing reference materials throughout the implementation process.

User Support Strategy

Providing effective user support is essential for sustaining the long-term success of STS8's adoption. Although the training program prepares employees to use the system before deployment, questions and technical issues will continue to arise as users begin incorporating dashboards and reports into their daily responsibilities. A comprehensive support strategy ensures that employees have access to timely assistance while reinforcing the knowledge gained during training (Stair & Reynolds, 2018). Providing ongoing support also encourages users to continue utilizing the new reporting environment instead of reverting to previous manual reporting processes or independently maintained spreadsheets.

Following the production rollout, STS8 should establish a tiered support structure that assigns responsibilities based on the complexity of the issue being reported. Routine questions related to dashboard navigation, report filtering, and common system functionality should be addressed through the IT Help Desk or designated departmental support personnel. More complex technical issues

involving data integration, report development, security, or system performance should be escalated to the BI development team for investigation and resolution. Establishing clear support responsibilities helps reduce response times while ensuring that issues are directed to personnel with the appropriate technical expertise. In addition to direct technical support, STS8 should maintain a centralized knowledge repository that provides employees with continuous access to training materials and support documentation. This repository should include user manuals, quick-reference guides, frequently asked questions, recorded training sessions, and troubleshooting documentation that employees can access independently as needed. Providing self-service support resources allows employees to resolve common questions without relying exclusively on technical staff while encouraging continuous learning throughout the organization. As new dashboards are introduced or existing reports are modified, these materials should be reviewed and updated to ensure they remain accurate and relevant.

User support should continue beyond resolving technical issues by promoting continuous improvement of the environment. Feedback collected through help desk requests, user surveys, refresher training sessions, and discussions with department managers should be evaluated to identify recurring challenges and opportunities for enhancement. This information can be used to improve training materials, refine dashboard functionality, and prioritize future system enhancements that better support business operations. Maintaining an ongoing feedback process reinforces the continuous improvement strategy discussed earlier while ensuring that the business intelligence environment continues to evolve alongside STS8's operational and strategic needs.

Continuous Improvement Program

The training program should be viewed as a living process that evolves alongside STS8 rather than as a one-time implementation activity. As organizational priorities change and new reporting requirements emerge, employees will require updated training to effectively utilize new dashboards and reports. The L&D team should periodically review training materials to ensure they remain aligned with current system functionality and organizational best practices. Regular evaluations of course content also help ensure that training reflects lessons learned from previous implementations and incorporates enhancements made to the business intelligence environment over time.

Continuous improvement should also include ongoing education for existing employees and structured onboarding for new hires. Refresher training sessions should be scheduled once a fiscal year to reinforce key concepts, introduce newly developed reporting capabilities, and address common challenges identified through help desk trends or employee feedback. New or transferring employees should complete role-based training as part of the organization's onboarding process so they develop the same foundational knowledge as existing staff. Establishing training as an ongoing organizational function helps STS8 maintain user proficiency, support long-term adoption of the business intelligence environment, and ensure that employees remain prepared to leverage organizational data as business needs continue to evolve.

Continuous improvement includes periodic governance reviews. These reviews help ensure the reporting environment continues supporting the organization's goals as business needs change over time. Rather than assuming dashboards and KPIs will remain effective indefinitely, leadership should periodically evaluate whether existing reports continue providing meaningful information for decision-making. Metrics that are no longer useful should be revised or removed. Likewise, new reporting requirements should be incorporated as operational priorities evolve. Executive leadership, data owners, and business stakeholders should participate in these reviews to ensure both organizational and operational perspectives are represented. The review process should also confirm that KPI definitions remain consistent across the organization and that dashboards continue reflecting current business

processes. Findings from each governance review should be documented and used to prioritize future reporting enhancements. Establishing governance reviews as a recurring activity helps ensure the reporting environment remains aligned with organizational strategy while continuing to provide reliable and actionable information.

Evaluation & Metrics

The previous sections of this project established a comprehensive framework for improving data governance and reporting within STS8 Logistics. Governance policies were developed to strengthen data quality, security, compliance, and stewardship practices, while reporting requirements and dashboard designs were created to transform operational data into meaningful information that supports decision-making. The implementation plan then outlined the activities necessary to deploy these governance and reporting initiatives throughout the organization. Although elements of performance measurement, reporting objectives, and governance oversight were discussed throughout these sections, they were presented within the context of each individual component. Bringing these elements together provides a clearer picture of how the overall governance program will be evaluated after implementation and how its success will be measured over time.

Key Performance Indicators

Implementation itself does not ensure that a governance initiative will achieve its intended goals and objectives. Once new processes and reporting tools are in place, STS8 Logistics must establish ways to measure performance and identify opportunities for improvement. Key performance indicators (KPIs) provide this structure by translating organizational objectives into measurable outcomes that allow leadership to evaluate whether governance and reporting initiatives are producing the desired results. They provide measurable evidence that can support decision-making and demonstrate whether organizational goals are being achieved. Effective performance measurement depends on identifying a limited number of meaningful indicators that are closely aligned with an organization's critical success factors instead of simply collecting large amounts of operational data (Parmenter, 2015). Establishing these measures during the evaluation phase helps ensure that STS8 can monitor governance effectiveness, evaluate reporting performance, and determine whether its data governance strategy is supporting operational improvements and better decision-making.

Data Governance KPIs

Data governance KPIs provide the foundation for evaluating whether the governance framework established for STS8 Logistics is operating as intended. These measures assess the effectiveness of the policies, standards, and controls implemented throughout the organization by monitoring key areas. As reporting capabilities developed in previous sections, Power BI dashboards and standardized reports also provide leadership with the ability to monitor many of these metrics in near real time, allowing emerging issues to be identified before they significantly affect business operations or reporting accuracy. While dashboards provide continuous visibility into governance performance, scheduled reports support more detailed reviews and trend analysis over longer reporting periods.

Establishing governance KPIs also provides a structured method for determining whether the STS8's governance strategy is producing measurable improvements. Governance metrics allow the organization to evaluate the effectiveness of their governance programs by measuring areas such as data quality, compliance, security, and accessibility while supporting ongoing adjustments as organizational needs evolve (Loshin, 2024). Table 9 summarizes the primary data governance KPIs that will be used to evaluate the success of the STS8 Logistics governance program. Each KPI identifies what will be measured, how performance will be calculated, the target level of performance, how often, and the organizational role responsible for monitoring the metric.

Table 9
Primary Data Governance Key Performance Indicators

KPI	Measurement	Calculation	Target	Frequency	Responsible Party
Data Accuracy Rate	% of validated records	$(\text{Accurate Records} \div \text{Total Records}) \times 100$	$\geq 98\%$	Daily Dashboard; Monthly review	Data Stewards
Data Completeness	Required fields populated	$(\text{Completed Fields} \div \text{Required Fields}) \times 100$	$\geq 99\%$	Daily Dashboard; Monthly review	Data Stewards
Data Consistency	Matching values across systems	$(\text{Matching Records} \div \text{Sampled Records}) \times 100$	$\geq 98\%$	Weekly	Data Stewards
Policy Compliance	Governance policy adherence	$(\text{Compliant Audits} \div \text{Total Audits}) \times 100$	$\geq 95\%$	Quarterly	Governance Committee
Security Incident Frequency	Unauthorized access/security events	Incidents per Month	$< 2/\text{month}$	Continuous monitoring; Monthly review	IT Security
Audit Remediation Rate	Findings resolved	$(\text{Closed Findings} \div \text{Total Findings}) \times 100$	$\geq 95\%$	Quarterly	Governance Committee

Note. The KPIs presented in this table are used to evaluate the effectiveness of the STS8 Logistics data governance program following implementation.

The governance KPIs were selected because they measure the areas that have the greatest influence on the effectiveness of STS8 Logistics' data governance initiatives. These measures work together to provide leadership with a balanced view of data quality, compliance, security, and accountability. Since these values are directly aligned with organizational objectives and monitored consistently over time, the performance measures will provide STS8 with the greatest value (Parmenter, 2015). The target values established for each KPI represent realistic performance expectations that support continuous improvement. Metrics related to data accuracy, completeness, and consistency establish expectations for maintaining reliable information before it is used for reporting and decision-making. Compliance, security, and audit measures evaluate whether governance controls remain effective and whether identified issues are addressed in a timely manner. Because these measures are

continuously monitored through Power BI dashboards and reviewed through scheduled governance reports, leadership can identify emerging trends, investigate the underlying causes of performance changes, and implement corrective actions before issues significantly affect business operations or reporting reliability.

Responsibility for monitoring each KPI aligns with the governance structure established earlier in this project. Data stewards oversee data quality measures because they are responsible for maintaining the integrity of organizational information, while the Governance Committee evaluates compliance and audit activities to ensure governance objectives continue to be met. Security-related KPIs remain the responsibility of the IT Security Team due to its role in protecting organizational systems and responding to potential threats. Assigning ownership in this manner strengthens accountability while ensuring that governance activities remain aligned with the broader operational objectives of STS8 Logistics.

Reporting & Analytics KPIs

While data governance KPIs evaluate the quality, security, and management of organizational data, reporting and analytics KPIs measure how effectively that information is delivered and used throughout STS8. The reporting environment developed in previous sections was designed to provide stakeholders with timely access to operational information through interactive Power BI dashboards and standardized reports. Simply implementing these reporting tools, however, does not guarantee that they will improve decision-making or operational performance. According to DeLone and McLean (2003), measures such as system use, user satisfaction, and organizational benefits are key indicators for evaluating the success of an information system. Monitoring dashboard usage, report adoption, user satisfaction, and reporting efficiency provides measurable evidence that employees are actively using the reporting environment and that it is supporting organizational objectives. Table 10 summarizes the primary reporting and analytics KPIs that will be used to evaluate the effectiveness of the STS8 Logistics reporting environment. Each KPI identifies what will be measured, how performance will be calculated, the target level of performance, the monitoring frequency, and the organizational role responsible for evaluating the results.

Table 10

Primary Reporting & Analytics Key Performance Indicators

KPI	Measurement	Calculation	Target	Frequency	Responsible Party
Dashboard Usage Rate	Active dashboard users	$(\text{Active Users} \div \text{Total Users}) \times 100$	$\geq 90\%$	Weekly	BI Team
Report Adoption Rate	Departments regularly using reports	$(\text{Departments Using Reports} \div \text{Total Departments}) \times 100$	$\geq 90\%$	Monthly	BI Team
User Satisfaction	Survey score	Average Survey Rating	$\geq 4.5/5$	Quarterly	BI Team

Average Report Generation Time	Time to generate reports	Average Report Runtime	< 10 seconds	Continuous	IT
Decision-Making Cycle	Time from data request to decision	Average Elapsed Time	20% reduction	Quarterly	Executive Leadership
Manual Reporting Reduction	Manual reports eliminated	Reduction from Baseline	≥ 75%	Monthly	BI Team
Business Process Efficiency	Time savings from reporting	Reduction from Baseline	≥ 15%	Quarterly	Operations

Note. The KPIs presented in this table are used to evaluate the effectiveness of the STS8 Logistics reporting and analytics program following implementation.

The reporting and analytics KPIs evaluate whether the reporting environment implemented at STS8 Logistics is delivering timely, accessible, and actionable information to decision-makers. Rather than measuring the quality of the underlying data, these KPIs focus on the effectiveness of the reporting tools themselves by evaluating user adoption, reporting efficiency, and overall system performance. Each KPI ensures that reporting performance can be evaluated consistently and that accountability is clearly established throughout the organization.

Several of the KPIs measure user adoption because even the most well-designed reporting environment provides limited value if employees do not actively use it to support operational and strategic decisions. Dashboard usage rates, report adoption rates, and user satisfaction scores provide insight into how effectively stakeholders are integrating Power BI dashboards and standardized reports into their daily responsibilities. Operational measures such as average report generation time, reductions in manual reporting effort, and improvements in decision-making cycles evaluate whether the reporting environment is increasing efficiency and reducing the time required to access meaningful business information. Performance targets were established using realistic expectations that encourage continuous improvement while remaining achievable during normal business operations.

Responsibility for monitoring these KPIs is shared among the Business Intelligence team, IT personnel, and organizational leadership. The Business Intelligence team oversees dashboard performance, report adoption, and user feedback to identify opportunities for improving reporting capabilities. IT personnel monitor system performance to ensure reports remain available and responsive, while executive leadership evaluates whether reporting initiatives are contributing to faster decision-making and improved operational performance. Together, these reporting and analytics KPIs provide measurable evidence that the reporting environment is supporting the organizational objectives established earlier in this project by improving visibility into business operations, increasing reporting efficiency, and enabling more informed decision-making.

Business Outcome KPIs

The previous Data Reporting section identified the operational metrics and dashboard designs needed to support decision-making across STS8 Logistics. Many of those dashboards were developed to

monitor key business activities such as fleet utilization, maintenance performance, transportation costs, customer service, and safety. While those dashboards provide leadership with visibility into day-to-day operations, business outcome KPIs establish the measurable criteria used to determine whether those reporting capabilities are contributing to the organization's strategic objectives. In other words, the dashboards answer what is happening within the organization, while business outcome KPIs help determine whether the organization is achieving the desired results. Because these measures are directly aligned with the reporting goals and objectives established earlier in this project, they provide a practical method for evaluating the overall effectiveness of the data governance and reporting implementation. Monitoring business outcomes such as operational efficiency, cost management, customer satisfaction, and risk reduction allows leadership to determine whether improvements in data quality and reporting are translating into measurable organizational performance. Table 11 summarizes the primary business outcome KPIs that will be used to evaluate the success of the STS8 Logistics governance and reporting initiative.

Table 11

Primary Business Outcome Key Performance Indicators

KPI	Measurement	Calculation	Target	Frequency	Responsible Party
On-Time Delivery Rate	Deliveries meeting schedule	(On-Time Deliveries ÷ Total Deliveries) X 100	≥ 95%	Daily	Operations Manager
Fleet Utilization	Fleet operating efficiency	(Active Vehicles ÷ Total Fleet) X 100	≥ 90%	Daily	Fleet Manager
Transportation Cost per Mile	Operating cost efficiency	Total Transportation Cost ÷ Miles Driven	5% reduction	Monthly	Finance
Customer Satisfaction	Customer experience	Average Survey Score	≥ 4.5/5	Quarterly	Customer Service
Maintenance Downtime	Vehicle availability	Total Downtime Hours	10% reduction	Weekly	Fleet Maintenance
Safety Incident Rate	Operational risk	Incidents per 100,000 Miles	15% reduction	Monthly	Safety & Compliance

Note. The KPIs presented in this table are used to evaluate the business outcomes resulting from the implementation of the STS8 Logistics data governance and reporting program.

The selected KPIs align directly with the reporting goals and objectives developed earlier in this project. Measures such as fleet utilization, transportation cost per mile, and maintenance downtime evaluate operational efficiency and asset performance, while customer satisfaction and on-time delivery rates assess the organization's ability to provide reliable transportation services. Risk-related measures, including safety incident rates, help determine whether governance practices and reporting insights are

contributing to safer operations and improved regulatory compliance. Collectively, these indicators provide a balanced view of organizational performance by measuring financial, operational, customer service, and risk management outcomes. Responsibility for monitoring these KPIs is distributed across the business functions that influence each area of performance. Operations managers oversee transportation and delivery metrics, fleet managers monitor asset utilization and maintenance performance, customer service leadership evaluates customer satisfaction measures, and safety personnel monitor operational risk indicators. By assigning ownership to the departments most directly responsible for each outcome, STS8 Logistics strengthens accountability while ensuring that performance improvements remain aligned with broader organizational objectives. When evaluated alongside the governance and reporting KPIs discussed in the previous sections, these business outcome measures provide a comprehensive assessment of the overall success of the organization's data governance and reporting implementation.

Collectively, these three categories of KPIs provide a comprehensive framework for evaluating the quality of organizational data, the effectiveness of the reporting environment, and the overall business value generated by the governance and reporting implementation.

Monitoring & Evaluation

The KPIs established in the previous section provide the framework for measuring the success of the STS8 Logistics data governance and reporting implementation. However, only defining performance measures is not enough to ensure long-term success. The company must also establish a structured process for monitoring those measures, evaluating results, and responding when performance does not meet established expectations. Ongoing monitoring and evaluation help ensure that the governance framework remains aligned with organizational objectives and adapts to changing business requirements and regulatory expectations.

Implementation should be viewed as the beginning of the governance lifecycle rather than its conclusion. As organizational processes, technologies, and reporting needs continue to evolve, governance activities must be reviewed regularly to verify that they continue to provide value and support informed decision-making. Data governance requires continuous monitoring and evaluation to maintain data quality, ensure policy compliance, and support continuous improvement throughout the organization (DAMA International, 2024). For STS8 Logistics, this ongoing evaluation process includes regularly scheduled performance reviews, governance committee meetings, compliance assessments, data quality audits, dashboard usage analytics, and system performance reviews. These activities provide leadership with the information needed to identify performance trends, address emerging issues, and implement corrective actions before they significantly affect organizational performance.

Performance Monitoring Plan

The monitoring frequencies established in Tables 1 through 3 define how often each KPI will be reviewed based on its importance, operational impact, and the need for timely intervention. While those tables identify the appropriate review schedule for individual performance measures, an effective monitoring plan also establishes what activities occur during each review period and how organizational

leaders use the results to evaluate the effectiveness of the governance and reporting program. Regular monitoring helps ensure that performance trends are identified early, governance objectives remain aligned with business priorities, and corrective actions can be implemented before issues significantly affect organizational performance.

To support this process, STS8 Logistics will follow a structured monitoring schedule that combines operational reviews with longer-term strategic evaluations. More frequent monitoring activities focus on identifying and resolving operational issues as they occur, while less frequent reviews evaluate broader organizational trends, governance effectiveness, and opportunities for continuous improvement. Table 12 summarizes the primary monitoring activities and their corresponding review schedule.

Table 12

Performance Monitoring Schedule

Review Frequency	Monitoring Activities	Evaluation Purpose
Weekly	Data quality checks, system health monitoring, security monitoring, exception reporting	Identify operational issues early, maintain data integrity, verify system availability, and address potential risks before they affect reporting or business operations.
Monthly	KPI performance reviews, dashboard usage analytics, report adoption tracking, manual reporting reduction analysis	Evaluate KPI performance against established targets, monitor user adoption, identify reporting trends, and assess operational efficiency improvements.
Quarterly	Governance committee meetings, compliance assessments, data quality audits, business performance evaluations	Review governance effectiveness, assess regulatory compliance, evaluate business outcomes, and prioritize corrective actions or improvement initiatives.
Annually	Strategic assessment, governance framework review, KPI evaluation, policy and reporting framework updates	Determine whether the governance program continues to support organizational objectives, evaluate long-term program effectiveness, and update governance policies, KPIs, and reporting strategies as business needs evolve.

Note. This monitoring schedule complements the KPI monitoring frequencies established in Tables 1 through 3 by defining the organizational review activities performed throughout the governance lifecycle. Monitoring results are used to identify trends, evaluate performance, and support continuous improvement initiatives.

Weekly and monthly monitoring activities are primarily operational in nature and emphasize the early identification of issues that could affect reporting accuracy, data quality, or system performance. Reviewing data quality metrics, system health, dashboard usage, and KPI performance on a regular basis allows leadership to identify negative trends before they become significant operational concerns. These reviews also provide insight into user adoption and reporting effectiveness, allowing the organization to determine whether employees are fully utilizing the Power BI dashboards and reporting tools developed during implementation. Quarterly and annual evaluations provide a broader assessment of the governance program by examining compliance, audit results, business performance, and the continued alignment of governance practices with organizational objectives. Governance committee meetings provide an opportunity to review KPI performance, discuss emerging risks, and prioritize improvement initiatives based on organizational needs. Annual strategic assessments evaluate the effectiveness of the

governance framework as a whole, ensuring that governance policies, reporting processes, and performance measures continue to support the long-term goals of STS8 Logistics as business requirements evolve. Although KPI performance provides quantitative evidence of program effectiveness, periodic user feedback, governance committee discussions, and operational reviews provide qualitative insight that helps leadership better understand the factors influencing performance.

By combining operational reviews with periodic strategic assessments, the organization can identify emerging issues, evaluate performance trends, and verify that governance activities remain aligned with business objectives. Frequent monitoring activities focus on maintaining day-to-day operations, while longer-term evaluations provide leadership with opportunities to assess the overall effectiveness of the governance framework and reporting environment.

Governance Reviews

Routine monitoring activities provide insight into individual KPIs, but governance reviews allow leadership to evaluate how those performance measures collectively reflect the effectiveness of the data governance and reporting program. These reviews shift the focus from day-to-day operations to a broader assessment of whether the organization's governance framework continues to support the strategic objectives established for STS8 Logistics.

Governance reviews are conducted primarily during the quarterly and annual review cycles. During these meetings, the Governance Committee examines KPI performance alongside compliance assessment results, data quality audits, dashboard usage trends, and other operational measures. Looking at these results together provides context that individual metrics cannot provide on their own. For example, declining dashboard adoption may be accompanied by lower user satisfaction scores or slower decision-making cycles, helping leadership better understand the underlying issue. Similarly, recurring data quality concerns may indicate that governance policies or operational procedures need additional attention. Reviewing organizational performance from this broader perspective allows leadership to identify trends, recognize opportunities for improvement, and determine whether the governance program continues to provide value to the organization.

Governance reviews also represent the point at which performance results become organizational decisions. When KPI trends consistently fall below established targets, leadership determines whether additional investigation is necessary and what corrective actions should be taken. Taking timely corrective action allows the organization to resolve underlying issues, strengthen governance practices, and support the continuous improvement necessary for long-term organizational success.

Corrective Action Process

Effective quality management depends on structured improvement processes that identify performance gaps and implement actions that lead to sustainable organizational improvement (Evans & Lindsay, 2020). This means that identifying performance deficiencies is only valuable if the organization responds appropriately. When governance reviews determine that KPI performance has consistently fallen below established targets, STS8 Logistics will initiate a structured corrective action process to

restore performance and reduce the likelihood of recurring issues. Corrective actions should be systematic rather than reactive, ensuring that organizational resources are focused on resolving the underlying problem instead of temporarily addressing its symptoms.

The first stage of the corrective action process is conducting a root cause analysis. Although KPI trends identify where performance has declined, additional investigation is necessary to determine why the issue occurred. Leadership should review operational data, examine dashboard trends, evaluate audit findings, and gather input from employees responsible for the affected business processes. This investigation helps determine whether declining performance is the result of data quality issues, ineffective business processes, insufficient employee training, technology limitations, or weaknesses within the governance framework. Understanding the underlying cause allows corrective actions to address the source of the problem rather than its immediate effects (Evans & Lindsay, 2020).

After the root cause has been identified, corrective actions should be selected based on the nature of the performance issue. For example, declining dashboard adoption may indicate that employees require additional training or that reports should be redesigned to better support operational decision-making. Data quality issues may require revised validation rules, changes to data entry procedures, or additional oversight from Data Stewards. If compliance assessments identify recurring policy violations, governance policies may need to be updated, communicated more effectively, or supported through additional monitoring activities. Selecting corrective actions that directly address the identified cause increases the likelihood that performance improvements will be sustained over time. Each corrective action should be documented to establish accountability and provide a record of organizational improvements. Documentation should identify the KPI or performance issue that triggered the investigation, summarize the results of the root cause analysis, describe the corrective action being implemented, assign responsibility for completing the action, and establish a reasonable implementation timeline. Maintaining this information creates an audit trail that supports future governance reviews while allowing leadership to monitor the status of improvement initiatives. It also provides valuable organizational knowledge that can be referenced if similar issues occur in the future.

Corrective actions should not be considered complete once changes have been implemented. Leadership should continue monitoring the affected KPIs to determine whether performance has improved and whether the corrective action successfully resolved the underlying issue. If KPI performance remains below established targets, additional investigation and corrective measures may be necessary. Parmenter (2015) emphasized that effective performance management requires organizations to regularly evaluate KPI results and use those findings to support ongoing improvement. By verifying the effectiveness of corrective actions and continuing to evaluate organizational performance, STS8 Logistics can strengthen its governance program and ensure that improvements produce measurable, long-term value.

Continuous Evaluation Strategy

Continuous evaluation ensures that the STS8 data governance and reporting program remains effective as organizational priorities and business requirements continue to evolve. While corrective actions address specific performance issues as they arise, continuous evaluation takes a broader view by

regularly assessing whether the governance framework continues to support the organization's strategic objectives. This ongoing approach recognizes that governance is not a one-time initiative but a continuous process that must adapt to changes. As organizations grow and their information needs change, governance practices should be reviewed and refined to ensure they continue delivering value (DAMA International, 2017).

As the organization grows, leadership should periodically evaluate whether existing KPIs, performance targets, governance policies, and reporting processes continue to provide meaningful measures of organizational success. Business priorities that are important during implementation may become less relevant over time, while new operational challenges or regulatory requirements may require additional performance measures or revised governance practices. Periodic reviews of the governance framework ensure that performance expectations remain realistic, reporting capabilities continue to meet stakeholder needs, and governance activities evolve alongside the organization rather than remaining static. Continuous evaluation also provides leadership with an opportunity to measure the long-term effectiveness of previously implemented improvements. By comparing current KPI performance with historical results, STS8 can determine whether corrective actions have produced sustainable improvements and whether additional refinements are necessary. This ongoing cycle of monitoring, evaluation, improvement, and reassessment allows the organization to continually strengthen its governance program while maintaining alignment with strategic business objectives and supporting informed decision-making across the enterprise.

Through regular KPI reviews, governance assessments, corrective action tracking, and periodic evaluation of governance objectives, leadership can determine whether the data governance and reporting implementation continues to improve operational performance while supporting the long-term strategic goals of STS8 Logistics.

Feedback Mechanism

Throughout this project, stakeholder feedback has been incorporated into nearly every stage of the proposed governance and reporting implementation. Feedback influenced the identification of reporting requirements, guided dashboard design decisions, supported governance oversight, informed change management activities, and shaped employee training and adoption strategies. Although these activities were presented within the context of their respective project phases, bringing them together establishes a formal feedback mechanism that ensures stakeholder input continues to support the governance program after implementation. While KPIs and performance monitoring provide measurable evidence of organizational performance, they do not always capture how users experience the governance and reporting environment in their daily responsibilities. Establishing structured feedback mechanisms allows STS8 Logistics to gather perspectives from the employees and stakeholders who rely on these systems to perform their work. According to Petter et al. (2013), user satisfaction and system use are important indicators of information system success because they provide organizations with insight into how effectively systems support user needs and organizational objectives. By integrating stakeholder feedback into the governance lifecycle, STS8 Logistics can identify opportunities for

improvement, prioritize enhancement initiatives, and ensure that its governance and reporting solutions continue to evolve alongside changing business requirements.

User Feedback Collection

A comprehensive feedback mechanism depends on collecting input through multiple channels. Different feedback mechanisms provide different perspectives on the effectiveness of the data governance and reporting program. While structured surveys and feedback forms generate measurable data that can be analyzed over time, interviews, focus groups, and stakeholder meetings provide additional context that helps explain why users are experiencing particular challenges or identifying opportunities for improvement. Using multiple collection methods also allows STS8 Logistics to gather feedback from employees with different roles and levels of interaction with the organization's governance and reporting environment.

Effective reporting solutions should also support the needs of the people who rely on them to make decisions. Collecting feedback through a variety of mechanisms allows STS8 Logistics to evaluate both the technical performance of its reporting environment and the overall user experience. Table 13 summarizes the primary feedback mechanisms that will be used to gather stakeholder input and support the continuous improvement of the organization's data governance and reporting program.

Table 13
User Feedback Collection Mechanisms

Feedback Mechanism	Primary Purpose	How It is Conducted	Typical Output
User Surveys	Measure user satisfaction and overall experience	Standardized electronic surveys distributed after implementation and periodically thereafter	Satisfaction scores, ease-of-use ratings, enhancement requests
Interviews	Gather detailed feedback on reporting effectiveness and business needs	One-on-one or small-group discussions with managers and key stakeholders	Operational concerns, reporting requirements, improvement recommendations
Focus Groups	Evaluate new dashboards, reports, or governance initiatives	Facilitated discussions with representative user groups	Usability feedback, feature suggestions, adoption barriers
Embedded Feedback Forms	Collect immediate feedback while users interact with Power BI dashboards	Feedback button or electronic form available within reporting tools	Dashboard-specific issues, usability recommendations
Help Desk Ticket Analysis	Identify recurring technical and operational issues	Review and categorize support requests submitted to IT	System issues, training gaps, technical improvement opportunities
Stakeholder Review Meetings	Review governance performance and strategic alignment	Quarterly governance committee and leadership meetings	Strategic recommendations, policy revisions, governance priorities
Suggestion Portal	Encourage continuous employee participation	Online portal where users can submit ideas throughout the year	Process improvements, reporting enhancements, innovation ideas

Note. Multiple feedback mechanisms provide both quantitative and qualitative information that supports continuous improvement of the STS8 Logistics data governance and reporting program.

Selecting appropriate feedback mechanisms is only one component of an effective feedback program. Equally important is ensuring that feedback is collected from stakeholders representing different levels of the organization, each of whom interacts with the data governance and reporting environment from a unique perspective. Executive leadership, operational managers, technical teams, and end users each evaluate different aspects of the governance program, allowing STS8 Logistics to develop a more complete understanding of organizational performance and opportunities for improvement. Table 14 identifies the primary stakeholder groups that will participate in the feedback process and the feedback mechanisms best suited to gathering input from each group.

Table 14

Primary Stakeholder Feedback Sources

Stakeholder Group	Primary Feedback Mechanisms	Primary Feedback Focus	Collection Frequency
Executive Leadership	Executive interviews, governance committee meetings	Strategic alignment, business value, organizational priorities, ROI	Quarterly and annually
Governance Committee	Governance review meetings, compliance reviews	Governance effectiveness, policy compliance, KPI performance, improvement initiatives	Quarterly
Department Managers	Interviews, stakeholder review meetings, suggestion portal	Operational reporting effectiveness, business process improvements, decision support	Quarterly
Data Stewards	Governance meetings, data quality reviews	Data quality, metadata, validation standards	Monthly
Dashboard Users / End Users	User surveys, embedded feedback forms, help desk tickets	Dashboard usability, report accuracy, ease of use, enhancement requests	Quarterly and ongoing
Business Intelligence (BI) Team	Stakeholder review meetings, user feedback analysis, help desk trends	Dashboard enhancements, report optimization, visualization effectiveness, report performance	Monthly
Information Technology (IT)	Help desk ticket analysis, system performance reviews, incident reports	System availability, infrastructure performance, technical support, application reliability	Ongoing and monthly
Information Security Team	Security monitoring, incident reviews, compliance assessments	Access controls, cybersecurity risks, policy compliance, security improvements	Ongoing and quarterly
Training and Change Management Team	Post-training evaluations, user surveys, focus groups	Training effectiveness, user adoption, knowledge gaps, additional training needs	Following training sessions and quarterly

Note. Stakeholders participate in the feedback process using mechanisms most appropriate to their organizational role.

The stakeholder groups identified in Table 14 represent the collaborative governance structure established throughout this proposal. By collecting feedback from leadership, operational users, governance personnel, and technical teams, STS8 Logistics can evaluate the effectiveness of its governance and reporting environment from multiple organizational perspectives. This collaborative approach reduces the risk of improvements being driven by a single department while ensuring that enhancement initiatives remain aligned with both operational needs and strategic business objectives.

Feedback Analysis & Prioritization

Collecting feedback from a variety of stakeholders provides valuable insight into the effectiveness of the data governance and reporting program, but meaningful improvements depend on how that information is evaluated and translated into action. STS8 Logistics will use a structured review process to analyze both quantitative and qualitative feedback, identify recurring trends, distinguish isolated concerns from organization-wide issues, and determine which recommendations should be implemented. A structured evaluation process helps ensure that improvement initiatives remain aligned with organizational objectives rather than being driven by individual preferences or isolated requests (Goetsch & Davis, 2021).

The analysis process begins by consolidating information collected through the feedback mechanisms presented in Table 13 and the stakeholder groups identified in Table 14. Quantitative measures, including user satisfaction scores, dashboard effectiveness ratings, report adoption rates, training assessment results, and support service metrics, provide measurable evidence of how effectively the governance and reporting environment is performing over time. Trend analysis allows leadership to compare current results with previous reporting periods, identify recurring performance issues, and evaluate whether previously implemented improvements have produced measurable benefits. Qualitative feedback collected through interviews, focus groups, governance meetings, help desk tickets, and suggestion portals complements these measures by providing context that numerical data alone cannot provide. Reviewing both forms of feedback together enables leadership to understand not only what performance trends are occurring, but also why they are occurring and how they affect different areas of the organization.

Following analysis, individual comments and recommendations should be organized into common themes before improvement initiatives are developed. Similar suggestions submitted by different stakeholder groups often indicate broader organizational concerns that warrant additional attention, whereas isolated requests may represent department-specific preferences rather than organization-wide needs. Feedback should also be evaluated alongside KPI performance, governance review findings, compliance assessments, and operational reporting metrics to determine whether reported concerns are supported by objective performance data. Integrating multiple sources of information in this manner allows STS8 Logistics to develop improvement recommendations based on evidence instead of isolated observations while ensuring that proposed changes remain consistent with established governance objectives (Project Management Institute [PMI], 2025). However, not every recommendation can or should be implemented immediately. Improvement initiatives should be prioritized according to their potential impact on business operations, governance objectives,

organizational risk, implementation complexity, and available resources. Recommendations involving data quality deficiencies, cybersecurity risks, regulatory compliance, or reporting accuracy should receive the highest priority because of their potential effect on organizational performance and decision-making. Operational improvements, such as dashboard enhancements, workflow optimization, or reporting efficiency gains, should be evaluated based on the value they provide relative to the effort required for implementation. Lower-priority enhancement requests, including cosmetic dashboard modifications or individual user preferences, may be incorporated into future development cycles after higher-impact initiatives have been completed. Establishing consistent prioritization criteria allows organizational resources to be directed toward improvements that provide the greatest strategic and operational value while maintaining transparency throughout the decision-making process (PMI, 2025).

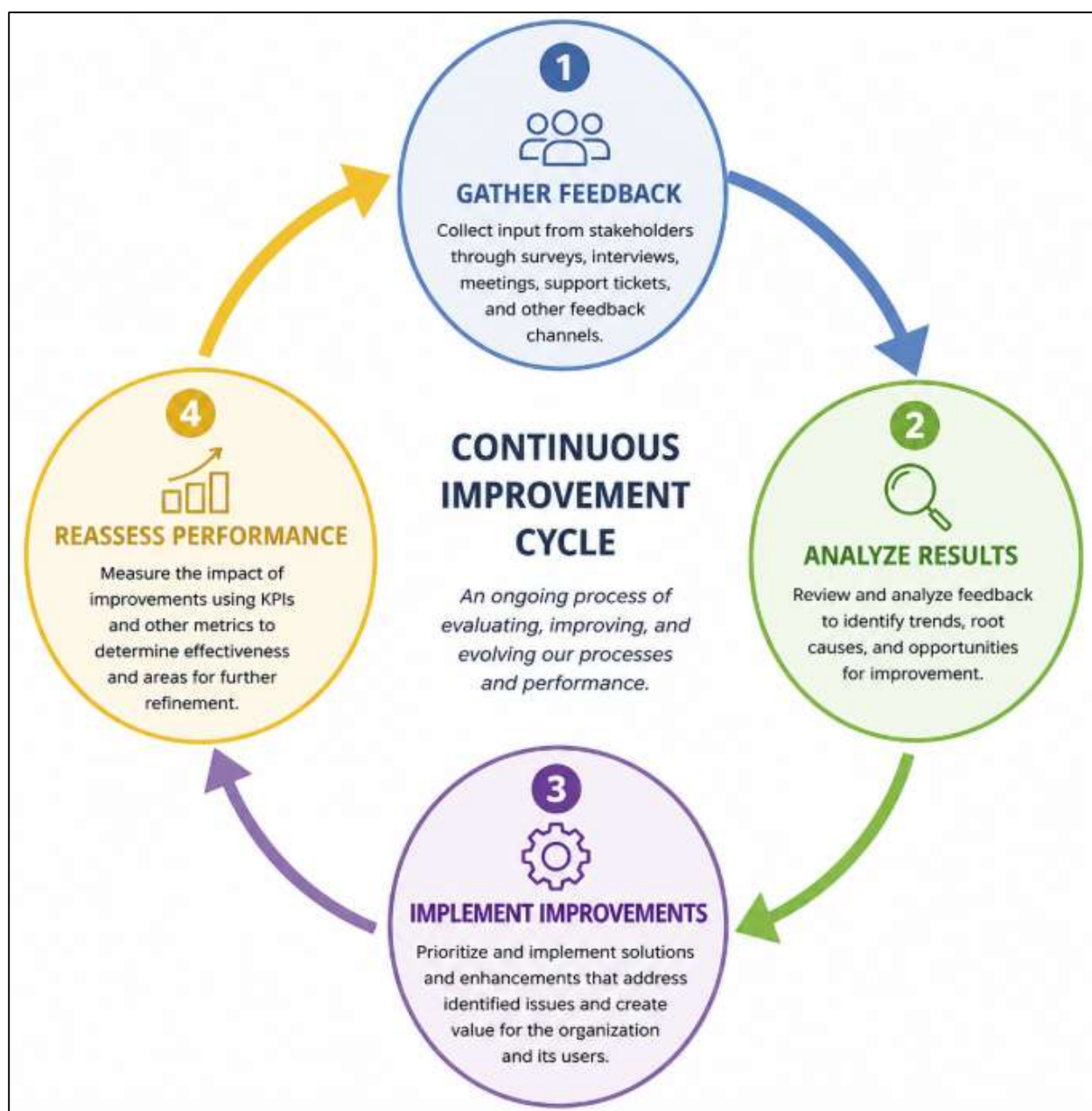
The Governance Committee, BI team, IT personnel, executive leadership, and other stakeholders each contribute to the prioritization process by evaluating recommendations from their respective areas of responsibility. Technical teams assess implementation feasibility and resource requirements, governance leaders evaluate policy implications and organizational risk, and executive leadership determines whether proposed improvements support the long-term strategic objectives of STS8 Logistics. This review process ensures that improvement decisions reflect both operational realities and organizational priorities while maintaining accountability throughout the governance program. By consistently analyzing stakeholder feedback, validating recommendations with performance data, and prioritizing initiatives according to organizational value, STS8 Logistics can make informed decisions that strengthen its governance framework and reporting environment while supporting continuous improvement.

Continuous Improvement Process

Goetsch and Davis (2021) emphasized that continuous improvement depends on regularly evaluating organizational performance, implementing evidence-based improvements, and reassessing results to verify that meaningful progress has been achieved. The purpose of collecting, analyzing, and prioritizing stakeholder feedback is to support a continuous cycle of organizational improvement. Once improvement initiatives have been approved, STS8 Logistics will incorporate them into the organization's governance, reporting, and operational planning processes according to their established priority and implementation timeline. Completed improvements should then be evaluated using the KPI framework, monitoring activities, and governance reviews established earlier in this project to determine whether the desired outcomes have been achieved. Figure 11 illustrates how stakeholder feedback progresses through each stage of this continuous improvement cycle, from collecting feedback and analyzing results to implementing improvements and reassessing organizational performance. This cyclical approach ensures that stakeholder feedback becomes an ongoing component of organizational decision-making rather than a one-time evaluation activity.

Figure 11

Continuous Improvement Process



Note. The figure illustrates the continuous improvement cycle used by STS8 Logistics to support its data governance and reporting program.

An effective feedback process also requires transparent communication with the stakeholders who provide input. Users should be informed when significant improvements are implemented through governance committee updates, departmental meetings, training sessions, release notes, and organizational communications. Communicating implemented changes demonstrates that stakeholder feedback is valued while increasing employee engagement and encouraging continued participation in future feedback activities. When appropriate, updated training materials and user documentation should accompany significant reporting or governance changes to ensure employees understand new processes and reporting capabilities. Maintaining open communication throughout the implementation process

strengthens user adoption and helps reinforce confidence in the organization's governance and reporting program.

Continuous improvement is not defined by the completion of individual enhancement projects but by the organization's ability to continually adapt as business requirements evolve. After improvements have been implemented and communicated, STS8 Logistics will continue monitoring KPI performance, collecting stakeholder feedback, evaluating governance effectiveness, and identifying new opportunities for improvement. As illustrated in Figure 11, this ongoing cycle of feedback collection, analysis, prioritization, implementation, communication, and reassessment ensures that the governance and reporting environment continues to evolve alongside organizational objectives. By embedding continuous improvement throughout the governance lifecycle, STS8 Logistics can sustain long-term user adoption, maintain effective data governance practices, and continue delivering meaningful business value through its analytics and reporting capabilities.

Conclusion

The purpose of this project was to develop a comprehensive data governance framework that supports accurate reporting and long-term organizational performance at STS8 Logistics. As the organization continues to grow, the volume and complexity of operational data generated across business units will continue to increase. Without consistent governance practices, these data assets become more difficult to manage and less reliable for business intelligence and strategic planning. This project demonstrated that effective data governance extends beyond regulatory compliance or data quality initiatives. It establishes the organizational structure, policies, and processes necessary to ensure information remains accurate, secure, consistent, and accessible throughout the organization.

The governance framework was designed to address these challenges by integrating governance principles into every stage of the information lifecycle. The framework established clearly defined governance roles, standardized policies, and measurable performance indicators to strengthen accountability and improve data quality. These governance activities support a consistent foundation for reporting and helps ensure organizational information is managed according to established standards. This project emphasized governance as an ongoing organizational capability that requires continuous monitoring, stewardship, and periodic review as business requirements evolve.

Building upon this governance framework, the project developed a business intelligence reporting environment designed to provide meaningful information to stakeholders throughout STS8. Reporting requirements were tailored to the needs of executive leadership, operational managers, fleet maintenance personnel, safety and compliance teams, and customer service representatives. Rather than presenting the same information to every user, each dashboard was designed to support the specific decisions made by its intended audience. The reporting environment focused on transforming governed data into actionable information that supports both operational and strategic decision-making. Long-term sustainability was also an important consideration throughout the reporting design. Dashboard ownership, standardized data refresh schedules, and established governance procedures help ensure reports remain accurate, relevant, and aligned with organizational objectives as business needs evolve.

Successful implementation depends on more than deploying new technology. For this reason, the project presented a structured implementation plan that addresses both the technical and organizational aspects of implementation. A hybrid project management methodology was selected

because it provides the structure needed for governance and infrastructure development while allowing dashboard development to evolve through iterative feedback. The implementation strategy also emphasized organizational change management and stakeholder engagement. Role-based training, ongoing user support, and continuous communication were incorporated to encourage adoption across the organization. Clearly defined responsibilities for both business and technical teams further strengthen accountability throughout the implementation process and support the long-term success of the governance program.

The project also established a comprehensive evaluation framework to measure governance maturity, reporting effectiveness, and organizational performance over time. Three categories of performance indicators were developed to support this effort. Data governance KPIs measure the quality and reliability of organizational information. Reporting and analytics KPIs evaluate user adoption and reporting performance. Business outcome KPIs determine whether governance and reporting initiatives contribute to meaningful operational improvements. Regular monitoring, stakeholder feedback, and governance reviews provide opportunities to identify issues before they become larger organizational problems. Corrective actions and continuous improvement activities help ensure the governance program continues adapting as business priorities change. Together, these components demonstrate that effective data governance is not simply a technology initiative. It is an organizational strategy that strengthens decision-making, improves operational visibility, and establishes a foundation for continuous improvement throughout STS8 Logistics.

Recommendations

The governance framework developed throughout this project provides STS8 Logistics with a strong foundation for managing organizational data and supporting BI. However, governance should never be viewed as a completed initiative. Organizations continue to grow, operational priorities change, and technology continues to evolve. Governance programs must evolve alongside those changes to remain effective. Continuous evaluation allows organizations to identify opportunities for improvement before existing processes become limitations. Policies should be reviewed regularly to ensure they continue supporting business objectives. Reporting requirements should also be reassessed as leadership priorities evolve. Viewing governance as an ongoing organizational capability encourages continuous improvement and positions STS8 to respond more effectively to future operational challenges.

Continuing Governance Maturity

As STS8 continues growing, leadership should periodically evaluate the overall maturity of the organization's governance program. The governance framework proposed throughout this project was designed to support the organization's current operating environment. Over time, however, additional business units, new technologies, and expanding reporting requirements may introduce complexities that require more advanced governance capabilities.

One area that should receive periodic evaluation is the organization's master data management strategy. Different implementation models provide different levels of data consistency, operational complexity, and organizational investment. Registry-based environments often provide an effective balance for organizations that need improved reporting consistency while allowing operational systems to continue functioning independently. Because master records remain within the source systems,

implementation can often be accomplished without fundamentally changing existing business processes (Robb, 2023). As governance practices mature, STS8 may eventually benefit from a coexistence model. Under this approach, a trusted "golden record" is maintained within the master data management environment while connected systems remain synchronized as updates occur. This creates greater consistency across the organization because validated information is distributed back to operational systems rather than existing independently within each application. Although coexistence provides stronger control over enterprise data, it also requires greater coordination among business units and significantly more governance oversight (Gils, 2020).

STS8 should periodically evaluate whether its existing governance architecture continues supporting organizational needs. As reporting requirements become more sophisticated and operational complexity increases, leadership should determine whether advancing to a more mature governance model would provide meaningful business value. Making these evaluations proactively allows the organization to expand its governance capabilities in a controlled manner rather than reacting after existing processes become operational constraints.

Expand Report Capabilities

The reporting environment must also continue evolving as additional organizational data becomes available. The dashboards developed throughout this project provide meaningful insight into current operations, but they should represent the foundation of a larger BI strategy. Future reporting initiatives should continue expanding self-service business intelligence capabilities. Providing managers with governed access to trusted organizational data allows them to answer many routine business questions without relying on the Business Intelligence team to develop new reports. This approach improves responsiveness while allowing technical personnel to focus on more complex analytical initiatives. STS8's next step will be periodically evaluate the information being collected throughout the organization. Additional operational data may support forecasting techniques that are not currently possible. Historical transportation trends, maintenance records, customer activity, and operational performance metrics can become increasingly valuable as larger datasets become available. Expanding these data sources creates opportunities for predictive reporting that helps managers identify emerging trends before they affect business performance. The objective should not be to create additional dashboards simply because more data exists. Instead, reporting should continue evolving toward providing increasingly meaningful insights that support proactive decision-making.

Future Analytics & AI

As STS8 Logistics continues to mature its capabilities, the organization's governance program should also evolve to address emerging analytical technologies. The governance framework presented throughout this project focuses primarily on managing organizational data, reporting processes, and decision-making activities. However, many organizations are beginning to incorporate predictive analytics and artificial intelligence (AI) into their reporting environments. As these technologies become more common, governance responsibilities will extend beyond managing data alone. Organizations must also establish oversight for the analytical models that transform data into business recommendations.

Predictive analytics has the potential to provide significant value within the logistics industry. Instead of just reporting historical performance, predictive models can forecast future conditions and identify emerging trends before they affect operations. For example, STS8 could eventually use predictive analytics to estimate shipment delays, forecast customer demand, identify vehicles requiring preventative maintenance, or recommend route adjustments based on historical transportation patterns. These capabilities would allow managers to address potential issues before they negatively affect operational performance. However, as organizations become increasingly dependent on predictive models, they must also ensure that those models remain accurate, transparent, and appropriately governed.

One important governance principle for AI-assisted decision support is maintaining meaningful human oversight throughout the decision-making process. Predictive models should provide recommendations that assist managers rather than replace managerial judgment. Operational decisions often require consideration of business conditions, customer relationships, regulatory requirements, or other factors that may not be fully represented within historical data. A Human-in-the-Loop (HITL) approach helps ensure that managers review analytical recommendations before significant operational decisions are made. This approach allows employees to validate recommendations, consider additional operational context, and override automated outputs when necessary. Maintaining human oversight also strengthens organizational accountability because responsibility for important business decisions remains with organizational leadership rather than automated systems (McCormack & Bendeckache, 2024).

Transparency must remain a central component of future governance efforts. Decision-makers are more likely to trust analytical recommendations when they understand how those recommendations were produced. Governance policies should therefore require documentation describing the purpose of predictive models, the data used during development, key assumptions, and known limitations. Providing this information helps stakeholders evaluate whether analytical outputs remain appropriate for their intended use. Greater transparency also allows governance committees to identify situations where changing business conditions may reduce model reliability or require additional review.

Just as STS8 monitors data quality and reporting performance, future governance activities should also include ongoing monitoring of analytical models. Model performance should be reviewed periodically to verify that predictions remain accurate as operational conditions evolve. Changes in customer demand, transportation networks, fuel costs, seasonal shipping patterns, or regulatory requirements may reduce the effectiveness of models that previously produced reliable results. Governance reviews should therefore evaluate prediction accuracy, monitor for model drift, and determine whether retraining or recalibration is necessary. These reviews should become part of the organization's existing governance review cycle so that analytical models receive the same level of oversight as data quality, reporting, and security controls. Ethical considerations should also become part of STS8's governance program as AI capabilities expand. Although predictive analytics can improve operational efficiency, organizations must ensure that automated recommendations remain fair, consistent, and aligned with business objectives. Governance policies should establish expectations regarding appropriate AI usage, accountability for automated recommendations, and documentation of significant model changes. Clear governance standards help reduce unnecessary risk while promoting responsible adoption of emerging analytical technologies (Hoffman & Evans, 2021).

Although STS8 Logistics is not currently implementing AI-assisted decision support, establishing these governance principles today will better position the organization to adopt future analytical capabilities responsibly. Expanding governance ensures that predictive models remain accurate,

transparent, and aligned with organizational objectives. As BI technologies continue to evolve, a mature governance framework will allow STS8 to adopt new capabilities while maintaining the accountability, reliability, and trust that effective decision-making requires.

Scalability

Finally, the proposed governance framework was intentionally designed to support long-term organizational growth. As STS8 expands its transportation network, governance practices should expand alongside it. New terminals, distribution facilities, fleets, and business units should adopt the same governance standards rather than developing independent reporting practices. This helps maintain consistency throughout the organization regardless of where operational data originates. The framework should be flexible enough to accommodate new technologies, additional operational systems, and changing business requirements without requiring complete redesigning. Policies and standards should be reviewed as new systems are being introduced to ensure they continue supporting enterprise-wide consistency.

Designing for scalability reduces future implementation effort and lowers the risk that governance practices become fragmented as the organization grows. It also positions STS8 to continue leveraging governed data as a strategic organizational asset regardless of how the business evolves. By establishing standardized governance principles today, the organization creates a framework capable of supporting both current operations and future expansion.

Reflection

Developing this project presented several challenges that ultimately strengthened the final outcome. One of the most significant challenges involved working with the logistics dataset itself. The dataset was selected early in the project during the proposal stage, long before it was analyzed in detail while developing the reporting environment. As a result, portions of the governance framework and reporting strategy were initially developed with assumptions about the available data rather than a complete understanding of its contents. When the reporting section was later developed, it became apparent that some of those assumptions did not fully align with the dataset. For example, I initially expected the customer data table to contain more detailed information that would support additional reporting and analysis. After examining the data more closely, I found that some of the insights I had planned to develop were not supported by the available information. Additionally, because the dataset was created for demonstration purposes rather than representing a real operational environment, some values lacked the variation that would normally exist in production data. These limitations required portions of the project to be revised so that the reporting environment accurately reflected the data that was actually available. Although revisiting earlier sections required additional effort, the experience reinforced one of the most important lessons discussed throughout my coursework. Before developing reports, conducting analysis, or designing governance strategies, it is essential to fully understand the available data, its metadata, and its limitations. Data dictionaries, field definitions, relationships, and data quality should have been evaluated before analytical work begun. This project demonstrated why that practice is so important. A governance framework can only be as effective as the information it is

designed to manage. Developing a thorough understanding of the data at the beginning of a project reduces unnecessary revisions and leads to more meaningful analytical outcomes

Throughout this course, my understanding of data governance and business analytics evolved in a different way than I initially expected. Many of the individual concepts were not entirely new to me because they had been introduced in previous courses. Throughout the program, I have worked with governance frameworks, SQL, business intelligence, data visualization, project management, systems analysis, implementation planning, and business analytics in separate assignments. While each course focused on a specific area, those topics were often explored independently. This capstone was the first opportunity to integrate those concepts into a single project that followed the complete lifecycle of a BI initiative. This demonstrated how each component depends on the others. The governance framework established the foundation for reliable data. That foundation supported the reporting environment, which then influenced the implementation strategy and ultimately the evaluation framework used to measure organizational success. Seeing those relationships develop throughout one project provided a much better understanding of how business intelligence initiatives are planned and executed within an organization.

Looking back at the completed project, I believe that was the most valuable lesson. It reinforced that successful BI initiatives are not built through a single technology, framework, or analytical technique. They are developed by integrating multiple disciplines into one cohesive strategy that supports organizational decision-making from beginning to end. Having the opportunity to connect concepts from across the curriculum into a single capstone project provided a perspective that individual assignments could not. It demonstrated how the knowledge gained throughout the program contributes to solving complex business problems in a practical and organized manner.

References

- Ali, M. A., Mahmood, A., Zafar, U., & Nazim, M. (2021). The power of adkar change model in Innovative Technology Acceptance under the Moderating Effect of Culture and Open Innovation. *LogForum*, 17(4), 485–502. <https://doi-org.csuglobal.idm.oclc.org/10.17270/J.LOG.2021.623>
- Archer, S., & Kaufman, C. (2013). Accelerating outcomes with a hybrid approach within a waterfall environment. Project Management Institute. <https://www.pmi.org/learning/library/outcomes-hybrid-approach-waterfall-environment-5839>
- BARC. (2017). *The most common business intelligence problems*. BARC. <https://barc.com/business-intelligence-problems/>
- Biswal, A. (2025, February 26). *Power BI vs Tableau: key differences and comparisons*. Simplelearn. <https://www.simplilearn.com/tutorials/power-bi-tutorial/power-bi-vs-tableau#:~:text=Data%20Integration%3A%20Power%20BI%20is,based%20databases%20and%20web%20services>
- Buhler, K. (n.d.). *How to use Tableau with Power BI and Fabric*. Data Goblin. <https://data-goblins.com/power-bi/connect-tableau-to-power-bi>
- Camm, J. (2021). *Data visualization: exploring and explaining with data*. Cengage Learning US. <https://reader2.yuzu.com/books/9798214341378>
- Colorado State University Global. (2026). *Foundation of data governance and the DAMA wheel*. Canvas course material.
- DAMA International. (2024). *The DAMA guide to the data management body of knowledge (DAMA-DMBOK2R)* (2nd ed., revised). Technics Publications. <https://dama.org/dmbok2r-infographics/>
- Data Governance Institute. (n.d.). *DGI data governance framework*. The Data Governance Institute. <https://datagovernance.com/the-dgi-data-governance-framework/>
- Dionisio, C. S. (2021). *Risk metrics: risks and impediments* [Video]. LinkedIn Learning. <https://www.linkedin.com/learning/data-driven-project-management-project-metrics-that-matter/risk-metrics-risks-and-impediments?u=2245842>
- Edenfield, O. (2026). *Microsoft Power BI data analyst certification guide*. Packt Publishing. <https://reader2.yuzu.com/books/9781803241326>
- Evans, J. R., & Lindsay, W. M. (2020). *Managing for quality and performance excellence* (11th ed.). Cengage.
- Gils, B. V. (2020). *Data management: a gentle introduction*. Van Haren Publishing. <https://reader2.yuzu.com/books/9789401805551>
- Goetsch, D. L., & Davis, S. B. (2021). *Quality management for organizational excellence: introduction to total quality* (9th ed.). Pearson.

- Gupta, D. (2024, December 17). *How to train employees on new software*. Whatfix. <https://whatfix.com/blog/train-employees-on-new-software/>
- Hoffman, E., & Evans, H. (2021). *Towards a standard for identifying and managing bias in artificial intelligence* (NIST Interagency/Internal Report 8371). National Institute of Standards and Technology. <https://doi.org/10.6028/NIST.IR.8371>
- ISACA. (2019). *COBIT 2019 framework: governance and management objectives*. ISACA. <https://netmarket.oss.aliyuncs.com/df5c71cb-f91a-4bf8-85a6-991e1c2c0a3e.pdf>
- kfolis, paulinbar, meganbradley, eva-lucas, maggiesMSFT, alexbuckgit, mohitp930, BassemMounir, cdpark, lafaberMSFT, mihart, monaraya, KesemSharabi, Sujata994, davidiseminger, john-par, v-hearya, LukaszPawlowski-MS, DennisLee-DennisLee, v-kents. (2025, March 18). *Share Power BI reports and dashboards with coworkers and others*. Microsoft. <https://learn.microsoft.com/en-us/power-bi/collaborate-share/service-share-dashboards>
- Kotter, J. (2012). *Leading change* (Rev. ed.). Harvard Business Review. <https://irp-cdn.multiscreensite.com/6e5efd05/files/uploaded/Leading%20Change.pdf>
- Lewin, K. (1947). Frontiers in group dynamics: concept, method and reality in social science; social equilibria and social change. *Human Relations*, 1(1), 5–41. https://lchc.ucsd.edu/mca/Mail/xmcamail.2013_07.dir/pdfef83xvxgaM.pdf
- Loshin, D. (2024, April 30). *Data governance metrics: How to measure success*. DATAVERSITY. <https://www.dataversity.net/articles/data-governance-metrics-how-to-measure-success/>
- McCormack, L., & Bendeckache, M. (2024). *Ethical AI Governance: Methods for Evaluating Trustworthy AI*. ArXiv. <https://arxiv.org/abs/2409.07473>
- Miller, R., Chan, S. H. M., Whelan, H., & Gregório, J. (2025). A comparison of data quality frameworks: a review. *Big data and cognitive Computing*, 9(4), 93. <https://doi.org/10.3390/bdcc9040093>
- Mohd Hanif, A. H., Lew, X. Y., Ibrahim, M. H., & Wong Hooi Ten, D. (2024). Get started in data governance and data stewardship. *Open International Journal of Informatics*, 12(1), 12–25. <https://doi.org/10.11113/oiji2024.12n1.236>
- National Institute of Standards and Technology. (2020). *Security and privacy controls for information systems and organizations* (NIST Special Publication 800-53, Rev. 5). U.S. Department of Commerce. <https://doi.org/10.6028/NIST.SP.800-53r5>
- Parmenter, D. (2015). *Key performance indicators: developing, implementing, and using winning KPIs* (3rd ed.). Wiley. <https://learning-oreilly-com.csuglobal.idm.oclc.org/library/view/key-performance-indicators/9781119019831/c01.xhtml#c1>
- Petter, S., DeLone, W., & McLean, E. R. (2013). Information systems success: the quest for the independent variables. *Journal of Management Information Systems*, 29(4), 7–62. <https://doi-org.csuglobal.idm.oclc.org/10.2753/MIS0742-1222290401>

- Phillips, J., & Klein, J. D. (2023). *Change management: from theory to practice*. *TechTrends*, 67(1), 189–197. <https://doi.org/10.1007/s11528-022-00775-0>
- Project Management Institute PMI. (2025). *A guide to the project management body of knowledge (PMBOK® Guide) — Eighth edition and the standard for project management* (8th edition). Project Management Institute. <https://research-ebsco-com.csuglobal.idm.oclc.org/c/wczg5c/ebook-viewer/epub/3ysduzybf5?location=https%25253A%25252F%25252Fresearch-ebsco-com.csuglobal.idm.oclc.org%25252Fc%25252Fwczg5c%25252Fsearch%25252Fdetails%25252F3ysduzybf5%25253Flimiters%25253DFT%2525253AY%252526q%25253DA%25252Bguide%25252Bto%25252Bthe%25252Bproject%25252Bmanagement%25252Bbody%25252Bof%25252Bknowledg%25252B%25252528PMBOK%2525252C2%252525AE%25252BGuide%252526searchMode%25253Dall>
- Robb, D. (2023). *4 popular master data management implementation styles*. Datamation. <https://www.datamation.com/big-data/master-data-management-implementation-styles/>
- Rodriguez, Y. (2025). *Logistics operations database*. Kaggle. <https://www.kaggle.com/datasets/yogape/logistics-operations-database>
- Saunders, M. N. K., Lewis, P., & Thornhill, A. (2023). *Research methods for business students* (9th ed.). Pearson.
- Schwenk, J. (2025). *Proposal Project: Baker Tilly business intelligence systems architecture* [Unpublished student paper]. Colorado State University Global.
- Smallwood, R. (2014). *Information governance: concepts, strategies, and best practices*. Wiley.
- Stair, R. M., & Reynolds, G. W. (2018). *Fundamentals of information systems* (9th ed.). Cengage Learning.
- Tableau. (n.d.). *Tableau server data engine*. Salesforce. https://help.tableau.com/current/server/en-us/data_engine2_intro.htm
- Valacich, J. S., & George, J. F. (2020). *Modern systems analysis and design* (9th ed.). Pearson.
- Vilminko-Heikkinen, R., & Pekkola, S. (2017). Master data management and its organizational implementation: An ethnographical study within the public sector. *Journal of Enterprise Information Management*, 30(3), 454–475. <https://doi.org/10.1108/JEIM-07-2015-0070>
- Weill, P., Van den Berg, E., Birnbaum, J., & de Planta, M. (2024). *What's next: top performers are becoming real-time businesses* (Research Briefing No. XXIV-8). MIT Center for Information Systems Research. https://cisr.mit.edu/publication/2024_0801_RealTimeBusiness_WeillvanderBergBirnbaumdePlanta?utm_source=pressrelease&utm_medium=pr&utm_campaign=realtimebusiness

Wick, A. (2022). *Change-management skills for business analysts* [Video]. Linked Learning.
<https://www.linkedin.com/learning/business-analysis-foundations-competencies/change-management-skills-for-business-analysts?autoAdvance=false&u=2245842>

Yin, R. K. (2018). *Case study research and applications: Design and methods* (6th ed.). Sage Publications.

AI Use Statement

ChatGPT (OpenAI) was used to support brainstorming, sentence refinement, paragraph structure, and alignment of prior coursework with current assignment requirements. Prompts focused on strengthening transitions between sections, improving clarity of written explanations, evaluating where previous academic work could support new rubric requirements, and helping written work become less repetitive. All analysis, interpretation, conclusions, and final written content remain the author's own work.